



Sharif University of Technology
Department of Aerospace Engineering

B.Sc. Thesis

Field: Flight Dynamics and Control

Design and Development of a Flapping-Wing
Mechanism Inspired by Dragonfly

Alireza Alinejad – 400106592

Taha Alikhani – 400106621

Supervisor: Dr. Seyed Hossein Pourtakdoust

A report submitted in partial fulfilment of the requirements of
Sharif University of Technology for the degree of
Bachelor of Science in *Aerospace Engineering*

September 2025

In the Name of God

Sharif University of Technology
Department of Aerospace Engineering

B.Sc. Thesis

Title:

Design and Development of a Flapping-Wing Mechanism Inspired by Dragonfly

Authors:

Alireza Alinejad
Taha Alikhani

Examination Committee:

Supervisor: Dr. Seyed Hossein Pourtakdoust

Signature: _____

Examiner: Dr. Alireza Sharifi

Signature: _____

Date of Defense: September 17, 2025

Abstract

Design and Development of a Flapping-Wing Mechanism Inspired by Dragonfly

Flapping-Wing Micro Aerial Vehicles (FWMAVs), especially those designed as independent flapping-wing mechanisms, have attracted serious attention in applications such as search and rescue, environmental monitoring, and reconnaissance missions due to their ability to perform complex maneuvers, capability of takeoff and hovering in confined spaces, and relatively low energy consumption.

The goal of this project is the design and construction of an initial prototype of a dragonfly-inspired flapping-wing vehicle featuring a controllable four-wing mechanism, and its main focus is on the mechanical design of this mechanism. In this path, four different mechanisms were designed and through simulation and physical prototyping, each of which proceeded until facing functional challenges, and then with a focus on fixing the defects, the next mechanism was designed to finally reach the final mechanism with the desired performance in the intended dimensions and weight.

The main characteristic of this design cycle has been gradual and test-driven development; such that each mechanism has acted as a structural step for subsequent modifications. Initial results show that the current flapping-wing mechanism has been validated and its mechanical operation is performed correctly; however, the capability of full and stable flight is currently still under investigation.

This project demonstrates one approach to step-by-step design for FWMAVs and smooths the path for the development of small, controllable, and efficient flapping-wing systems.

Keywords: Flapping mechanism, Flapping-wing, Dragonfly-like, FWMAV, Ornithopter

Preface

Since the beginning of my entry into the field of aerospace engineering, I have had a special interest in small and precise systems. While the major attention of the aerospace community is focused on the development of large and complex products such as airplanes, satellite carriers, and missiles, I have found the most attraction in designing systems that operate in small dimensions but with high delicacy and precision. This interest led me to pursue a topic in my undergraduate project that is less seen in the mainstream of this field: designing a flapping-wing mechanism for nature-inspired micro air vehicles.

This project was an opportunity for me to both test my technical abilities in designing complex mechanisms and pursue my interest in the world of precise and lightweight systems. I hope that the experience gained from this path will be a step, however small, toward the development of modern flight technologies.

Alireza Alinejad
August 26, 2025

Acknowledgements

Here, I consider it necessary to sincerely thank all those who helped and accompanied me throughout this project. First of all, I am grateful to my supervisor, Dr. Seyed Hossein Pourtakdoust, who smoothed the path for the realization of this project with his valuable guidance and continuous support. I also thank Dr. Seyed Mohammad Sadegh Mousavi, a PhD student of Dr. Pourtakdoust, with whom I consulted and exchanged ideas at various stages of the design and development of the mechanisms.

Special thanks also go to Dr. Torkaman Asadi and Mr. Seyed Amirhossein Shafaei, who played an important role in completing the mechanism with their support in the field of 3D printing using Fused Deposition Modeling (FDM) of some required parts.

Finally, I am grateful with all my being to my dear family, and especially my noble father, who have always been my supporter and backer both spiritually and financially in providing various tools and parts. Without their support and companionship, the completion of this project would not have been possible.

Contents

List of Figures	viii
List of Tables	x
1 Introduction	1
1.1 Problem Statement and Project Importance	2
1.2 Objectives and Innovations	2
1.3 Report Content	2
2 Research Background	4
2.1 Brief Introduction to FWMAV and Applications	4
2.2 Review of Similar Research and Mechanisms	5
2.2.1 Microbat: Development of the first palm-sized electric flapping-wing (2001)	5
2.2.2 Optimization of flapping mechanism and design of biomimetic wings (2008)	5
2.2.3 Review of bird flight and its inspiration in MAV design (2012)	5
2.2.4 BionicOpter: Recreation of dragonfly flight by Festo (2014)	5
2.2.5 Development, design, construction, and testing of FWMAV (2016)	5
2.2.6 Design of a large-scale flapping-wing bird with long stable flight (2017)	5
2.2.7 Dove: Development of an autonomous biomimetic bird (2018)	6
2.2.8 Introduction of a novel mechanism with elliptical wing tip path (2019)	6
2.2.9 RoboBee: Free flight of an insect-scale flapping robot (2013–2019)	6
2.2.10 Bio-inspired paths and production of lightweight FWMAV (2020)	6
2.2.11 Design, aerodynamic analysis, and experimental testing (2021)	6
2.2.12 Kakuta’s flapping-wing projects: Flapper and dragonfly-like models (2011–2023)	6

2.3	Strengths and Weaknesses of Past Research	7
3	Fundamentals of Flapping-Wing Mechanism Design	8
3.1	Design Input Parameters	8
3.2	Aerodynamics and Flight Dynamics Principles in Micro Dimensions . .	9
3.2.1	Force Modeling on the Wing	10
3.2.2	Lift Distribution on the Wing	11
3.2.3	Relation Between Force Modeling and Lift Distribution	13
3.2.4	Extraction of Aerodynamic Characteristics Using Software	13
3.3	Mechanism Design Criteria	14
3.4	Design and Simulation Tools	15
4	Development of Mechanisms	16
4.1	Coil and Magnet Mechanism	16
4.1.1	Design and Performance	16
4.1.2	Results and Limitations	20
4.1.3	Potential Improvements for Future Work	23
4.2	Coreless Motor with Crank-Slider Mechanism	25
4.2.1	Design and Performance	26
4.2.2	Results and Limitations	30
4.2.3	Potential Improvements	34
4.3	Dove-Inspired Gear Mechanism	36
4.3.1	Reference System: The Original Dove Mechanism	37
4.3.2	Adaptation for Four-Wing Configuration	37
4.3.3	Estimated Parameters	39
4.3.4	Limitations and Development Status	39
4.3.5	Potential Future Improvements	41
4.4	BionicOpter-Inspired Mechanism	43
4.4.1	Reference System: Festo BionicOpter	43
4.4.2	Design and Performance	43
4.4.3	Simulation Results	49
4.4.4	Experimental Validation	51
4.4.5	Prototype Assessment	55
4.4.6	Potential Improvements	56

5	Prototyping and Implementation	58
5.1	Wing Design	58
5.1.1	Base Geometry	59
5.1.2	Materials and Fabrication	59
5.1.3	Construction Process	59
5.2	Motor Selection	60
5.2.1	Torque Calculation	60
5.2.2	Motor Constants	60
5.2.3	Selected Motor: T-Motor F15	61
5.3	Mass Budget and Final Assembly	61
5.4	Experimental Setup	61
6	Conclusion and Wrap-up	64
6.1	Key Project Achievements	65
6.2	Main Innovations	65
6.3	Suggestions for Future Works	66
6.3.1	Short-Term Improvements (Mechanical Refinement)	66
6.3.2	Medium-Term Goals (Flight Control)	66
6.3.3	Long-Term Research (Optimization)	66
	References	67
	Appendices	69
A	Motor Selection Data	69
A.1	Motor Comparison Table	69

List of Figures

3.1	(A) Shows how the spanwise bending creates a $-F_z$ component of the aerodynamic force which acts in $-z_4$ direction in the sensor frame. The centrifugal force acts in the sensor $+z_4$ direction. (B) Shows how the spanwise twist $\delta\psi_i$ at the i^{th} element contributes to the $-F_x$ component of aerodynamic force in the sensor frame [1].	10
3.2	Lift force distribution on the wing	12
4.1	CAD model of the coil and magnet mechanism	17
4.2	Simulation results of the coil-magnet mechanism: angular displacement, angular velocity, driving current, and electromagnetic torque versus time. The limited motion amplitude under realistic loads is evident.	22
4.3	Fabricated prototype of the coil and magnet mechanism	22
4.4	CAD model of the coreless motor mechanism with single-stage gear reduction and crank-slider linkage. Four identical assemblies are used, one for each wing.	25
4.5	Simulation results for one coreless motor crank-slider assembly at 7 Hz flapping frequency: (a) slider displacement, (b) wing angle, (c) hinge torque components, (d) slider forces, (e) crank torque, (f) motor current and voltage.	31
4.6	Fabricated prototype of the four-motor coreless mechanism	33
4.7	CAD model of the Dove-inspired gear mechanism adapted for four-wing configuration	36
4.8	Detailed design of the Dove-inspired mechanism showing gear train, rack-and-pinion, and four-bar linkage assemblies	39
4.9	Evolution of CAD models during development of the BionicOpter-inspired mechanism, showing iterative refinement of gear train and wing attachment designs	45
4.10	Final CAD model of the BionicOpter-inspired mechanism showing the central motor, three-stage gear train, eccentric shaft converters, and wing attachment points	46

4.11	Simulation results for the BionicOpter-inspired mechanism at 15 Hz: (a) slider displacement, (b) wing angle, (c) hinge torque components, (d) slider forces, (e) motor torque, (f) motor current and voltage	49
4.12	Frequency sweep analysis: (a) motor speed vs. flapping frequency, (b) peak motor current vs. frequency, (c) required voltage vs. frequency with feasibility indication, (d) voltage margin vs. frequency	51
4.13	Ometron VH300+ Laser Doppler Vibrometer (Model OL27) used for high-frequency wingbeat measurement	52
4.14	Fabricated prototype of Mechanism 4.4 (intermediate version) with reflective target label attached to the wing for laser vibrometer tracking .	52
4.15	Experimental data analysis: (a) raw optical sensor voltage showing amplitude increase in the max throttle zone (red shading), (b) band-pass filtered signal (1–40 Hz), (c) wingbeat frequency extracted via zero-crossing method, showing stable operation at ~20 Hz during maximum throttle	53
4.16	Final prototype (Mechanism 4.6) showing all gear stages with modules and tooth counts, servo motors for rear wing control, and balancing weights for vibration reduction	55
5.1	Fabricated wing prototype showing the carbon fiber spar structure and polymer membrane	58
5.2	Final assembled vehicle on precision scale showing a mass of 82.4 g (including balance weights)	62
5.3	Experimental test bench showing power supply, ESC, controller, and data acquisition interface	62
6.1	The final BionicOpter-inspired prototype (Mechanism 4.6) demonstrating synchronized flapping motion during testing	64

List of Tables

3.1	Estimated values for ornithopter weight	8
3.2	Input parameters of the ornithopter	9
3.3	Meanings and Values of Lift Force Distribution Formula Coefficients . .	12
4.1	Key parameters of the coil-magnet mechanism	18
4.2	Motor and gear parameters (per wing, ×4 total)	27
4.3	Wing geometry and dynamics parameters (per wing)	28
4.4	Frequency sweep analysis (per motor)	32
4.5	Reference parameters from the original Dove system [2]	37
4.6	Estimated parameters for the Dove-inspired mechanism vs. project targets	40
4.7	Comparison of original BionicOpter with project targets	44
4.8	T-Motor F15 KV6000 specifications	44
4.9	Gear specifications	46
4.10	Gear train configuration	47
4.11	Eccentric converter and wing parameters	47
4.12	Simulation results at 15 Hz flapping frequency (7.4 V supply)	50
4.13	Experimental frequency measurements at maximum throttle	54
4.14	Comparison of simulation and experimental results	54
5.1	Mass breakdown of the final prototype	61
A.1	Motor selection based on MRT	69

List of Abbreviations

AoA	Angle of Attack
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CFRP	Carbon Fiber Reinforced Polymer
ESC	Electronic Speed Controller
FDM	Fused Deposition Modeling
FWMAV	Flapping-Wing Micro Aerial Vehicle
IMU	Inertial Measurement Unit
MAV	Micro Aerial Vehicle
MEMS	Micro-Electro-Mechanical Systems
MRT	Motor Required Torque
PET	Polyethylene Terephthalate
PLA	Polylactic Acid
PWM	Pulse Width Modulation
TFN	Total Force Needed
VLM	Vortex Lattice Method
UVLM	Unsteady Vortex Lattice Method

Chapter 1

Introduction

Flapping-wing flight in birds, bats, and insects, due to high aerodynamic efficiency at low Reynolds numbers, has always been an inspiration for novel research in the field of Flapping-Wing Micro Aerial Vehicles (FWMAVs). Compared to propeller-driven or fixed-wing configurations at similar scales, flapping-wings offer capabilities such as high maneuverability, the possibility of flight in confined spaces, and the ability to perform stealthy flight [3].

Studies have shown that the flight mechanism of flapping-wing animals involves multiple degrees of freedom, active changes in wing shape, and complex oscillating motions, achieving which in artificial designs is challenging [3–5]. However, recent advancements in actuator technology, lightweight materials, and control systems have made it possible to construct flight models with stable performance. For instance, the Dove project at Northwestern Polytechnical University in China presented a biomimetic FWMAV capable of autonomous flight and practical missions such as real-time video transmission [2].

However, one of the main challenges in designing FWMAVs is the development of a mechanism that, while maintaining delicacy and lightness, can offer acceptable and stable performance in a real environment. Most research focuses on structural or aerodynamic analyses, but the process of gradual mechanism design—an approach involving design, testing, facing challenges, modification, and advancement to the next mechanism—has received less attention.

The importance of the present project lies in precisely focusing on this step-by-step approach: several different mechanisms were designed and tested, and each was improved after facing practical challenges and transferred to the next mechanism to finally reach a mechanism with desired performance in specific dimensions and weight. This approach is a structured and experimental model for the development of FWMAVs that can be functional in future missions such as reconnaissance, environmental monitoring, and operations in confined spaces.

1.1 Problem Statement and Project Importance

The main challenge in the development of FWMAVs is that flight in small dimensions has different rules compared to conventional flight. At scales below 100g and wingspans under 50cm, airflow is more unstable, and conventional systems like fixed wings or propellers do not have suitable efficiency. Therefore, achieving stable, lightweight, and efficient flight requires designing specific flapping-wing mechanisms that can simultaneously generate lift and thrust and allow for hovering and complex maneuvers.

Despite extensive efforts, many existing samples still face limitations such as high weight, low efficiency, or difficulty in performance repeatability [3, 6, 7]. In the meantime, designing a mechanism that, in addition to suitable efficiency, has simplicity of construction and assembly, mechanical reliability, and low weight, is a serious necessity. This project is formed based on this very need and, inspired by the flight capabilities of the dragonfly, has followed the path of gradual development of several different mechanisms. The significance of this work lies in demonstrating that iterative prototyping can identify failure modes early and guide design refinements for correction and upgrades, and finally led to a mechanism that can be the basis for designing future generations of small and efficient FWMAVs.

1.2 Objectives and Innovations

The main goal of this project is the design and development of a flapping-wing mechanism inspired by the dragonfly that can be a basis for building lightweight, efficient, and controllable FWMAVs. The main focus was on the mechanical design and step-by-step investigation of several different mechanisms to achieve a stable and operational mechanism through continuous testing and modification.

The project's innovation lies in adopting a cyclical and test-driven approach; each mechanism was not merely a final product but a stage for learning, identifying limitations, and transferring modifications to the next design. However, each mechanism, considering its strengths and weaknesses, also has the potential for development and conversion into an independent final product.

The final mechanism was designed inspired by dragonfly flight capabilities and with attention to the requirements of “low weight”, “simplicity of construction and assembly”, and “performance stability” 3.2. This project shows that the development of future FWMAVs is possible not by a single design, but through an iterative process of design, testing, and improvement.

1.3 Report Content

This report is organized in six chapters. In the first chapter, the introduction and generalities including the problem statement, project importance, objectives, and main innovations are explained. The second chapter is dedicated to the literature review and clarifies the position of this project by reviewing past studies, applications, and similar mechanisms in the field of micro flapping-wing vehicles. In the third chapter,

the fundamentals of flapping mechanism design and the principles of aerodynamics and flight dynamics in micro scale are examined, and design criteria are introduced.

In the fourth chapter, the process of development and evaluation of four different mechanisms including the Coil-Magnet mechanism, Coreless mechanism, Dove mechanism, and the mechanism inspired by Bionicopter is presented, and the strengths and weaknesses of each are analyzed. The fifth chapter is dedicated to the prototyping and implementation process, discussing topics such as wing design and construction, motor selection, estimation of the bird's final mass, and test equipment. Finally, the sixth chapter is the conclusion, which includes the key achievements of the project, main innovations, and suggestions for continuing the research and development path.

Chapter 2

Research Background

Flapping-Wing Micro Aerial Vehicles (FWMAVs) have attracted the attention of many researchers in recent years. These systems are designed inspired by the flight of birds and insects, and their goal is to achieve stable, lightweight, and efficient flight at small scales. To understand the position of the current project, a brief review of basic concepts, applications, and previous research is necessary.

2.1 Brief Introduction to FWMAV and Applications

Flapping-Wing Micro Air Vehicles (FWMAVs) are a new generation of lightweight flight systems designed inspired by birds, bats, and insects. At small scales, flight with fixed wings or propellers experiences reduced aerodynamic efficiency and faces challenges with flow separation at low Reynolds numbers; whereas flapping can provide the possibility of simultaneous generation of lift and thrust and guarantee flight stability and performance at low speeds.

From an application perspective, these vehicles have capabilities that make them superior for specific missions. Flight in enclosed and confined environments such as buildings or caves, performing reconnaissance and environmental monitoring operations, search and rescue in disaster-stricken areas, and even covert military missions are among the fields where these vehicles have a competitive advantage. Furthermore, their visual resemblance and flight pattern to real birds and insects allow them to operate inconspicuously in surveillance missions.

In the past two decades, numerous projects worldwide have addressed the development of FWMAVs. Some examples have been introduced only as technology demonstrators, but another group has reached the level of operational performance and has been able to execute missions such as stable flight, live video transmission, or complex maneuvers.

2.2 Review of Similar Research and Mechanisms

2.2.1 Microbat: Development of the first palm-sized electric flapping-wing (2001)

The Microbat project at the California Institute of Technology was one of the first efforts to build micro flapping-wing birds. This bird, weighing 12.5 grams, was able to perform short flights with supercapacitors (9 seconds) and longer flights with rechargeable batteries (42 seconds). The use of MEMS technology in wing design was a significant innovation of this project [8].

2.2.2 Optimization of flapping mechanism and design of biomimetic wings (2008)

Liu et al. [9] used optimization methods to investigate four-bar mechanisms and the design of biomimetic wings. The results showed that coordination between the mechanical actuator and wing structure plays a key role in reducing energy consumption and increasing flight stability.

2.2.3 Review of bird flight and its inspiration in MAV design (2012)

Gerdes et al. [3], in a comprehensive review, analyzed the aerodynamic principles of bird and insect flight. The importance of phenomena such as “clap and fling” and “wake capture” for lift generation at micro scales was highlighted, providing a scientific basis for designing new FWMAVs.

2.2.4 BionicOpter: Recreation of dragonfly flight by Festo (2014)

Festo, with the introduction of BionicOpter, implemented independent control of four dragonfly-like wings in an artificial system for the first time. This bird was capable of hovering, flying in various directions, and even flying backwards, although it was known more as a technology demonstration [4].

2.2.5 Development, design, construction, and testing of FWMAV (2016)

Smith [10] tested various mechanisms and wings in his research. He showed that the correct choice of actuator mechanism and wing geometry has a direct impact on aerodynamic efficiency and maneuverability of micro birds.

2.2.6 Design of a large-scale flapping-wing bird with long stable flight (2017)

Pan et al. [6] developed a flapping-wing bird with a wingspan of 1.1 meters and a weight of 455 grams that could perform continuous flight for more than 15 minutes. This project highlighted the importance of wing flexibility and its role in increasing flight stability.

2.2.7 Dove: Development of an autonomous biomimetic bird (2018)

The Dove project at NPU University in China was an example of an operational FWMAV. This bird could perform fully autonomous flight and transmit live images up to several kilometers. The use of flexible wings with passive deformation is considered its main innovation [2].

2.2.8 Introduction of a novel mechanism with elliptical wing tip path (2019)

Liu et al. [5] designed a mechanism that produced wing tip motion in an “ ∞ ” shaped elliptical path. This design established a balance between mechanical simplicity and aerodynamic efficiency, leading to an improvement in overall flight efficiency.

2.2.9 RoboBee: Free flight of an insect-scale flapping robot (2013–2019)

The RoboBee project at the Wyss Institute at Harvard University is one of the most prominent achievements in this field. This robot, weighing 80 mg and with a wingspan of 3 cm, was able to perform controlled flight using piezoelectric actuators and later untethered free flight. This achievement was a turning point in FWMAV development [11].

2.2.10 Bio-inspired paths and production of lightweight FWMAV (2020)

Lane et al. [7], focusing on the design and production of lightweight FWMAVs, examined various wing configurations (single-wing, bi-wing, and quad-wing). They showed that the location and angle of spars have a significant effect on efficiency and flight performance.

2.2.11 Design, aerodynamic analysis, and experimental testing (2021)

In this study, simulation data were compared with flight test results. Results showed that combining numerical and experimental methods can effectively increase the validity of FWMAV aerodynamic models [12].

2.2.12 Kakuta’s flapping-wing projects: Flapper and dragonfly-like models (2011–2023)

Kazuhiko Kakuta in Japan developed numerous flapping-wing birds, including the recreation of the Flaptter from the animation *Castle in the Sky* and the design of dragonfly-like flapping-wings. Models such as Flaptter8-19 and Dragonfly48 demonstrated stable flight and acrobatic maneuvers. A 1/4 cycle phase difference between the front and rear wings in these projects improved flight efficiency and stability [13, 14].

2.3 Strengths and Weaknesses of Past Research

A review of previous research shows that the development of micro flapping-wing birds has made significant progress in the last two decades. Among the most important strengths of these studies, the following can be mentioned:

- **Realization of stable flight at small scales:** Examples like Microbat and RoboBee showed that controlled flight is possible at masses ranging from 80mg (RoboBee) to 12.5g (Microbat) and more, using lightweight actuators and MEMS wings.
- **Innovation in flight mechanisms:** Projects like BionicOpter and Dove, by introducing independent four-wing motion or using passive wing deformation, demonstrated capabilities such as hovering, multi-directional flight, and live data transmission.
- **Advancement in manufacturing and materials:** The use of modern technologies like MEMS and lightweight composite materials allowed for weight reduction and efficiency increase.

Despite these achievements, common challenges and weaknesses still remain:

- **Complexity of construction and assembly:** Many examples have multi-motor or multi-degree-of-freedom mechanisms that make their repeatability and assembly difficult.
- **Limited flight capacity:** Most examples either have short-duration flights (like Microbat) or depend on an external power source (like early generations of RoboBee).
- **Limitation in payload:** Due to very low weight, the ability to carry larger sensors or batteries is limited.
- **Lack of a gradual design framework:** Most research has focused on analyzing or building a single sample and has less followed a staged and test-driven approach for mechanism evolution.

These strengths and weaknesses indicate that although the FWMAV development path has been illuminated, there is still a need for simpler, lighter, and gradual design approaches; a goal that the current project focuses on.

Chapter 3

Fundamentals of Flapping-Wing Mechanism Design

3.1 Design Input Parameters

The values related to the estimated weight of the ornithopter are presented in the table below.

Table 3.1: Estimated values for ornithopter weight

Name	Num.	Each Mass (g)	Total Mass (g)
Rechargeable Li Polymer Battery Pack LP402040 2S 7.4V 300mAh With Protection Circuit and Wires	1	12	12
PZ-15320 micro Servo motor	2	1.7	3.4
Brushless Motor (T-motor F15)	1	8	8
Arduino Pro Mini 5 - 12 V (5V model) / Wemos S2 mini	1	2	2
T-motor AIR 10A Brushless Speed Controller ESC	1	8	8
T-motor AIR 15A Brushless Speed Controller ESC	1	10	10
Gears + Plates	1	16.5	16.5
Carbon Fiber Rod (3mm) (12cm)	4	0.996	3.984
Wing Structure + Layer	4	3	12
Bearing	14	0.3	4.2
LED	2	0.5	1
GY-BMI160 6axis IMU	1	2	2
UVC1-AR0522-C-S85-IR (PCB Camera)	1	18	18
Total Full Payload			≈ 93.1

Also, the input parameter values of the ornithopter are presented below.

Table 3.2: Input parameters of the ornithopter

Parameters	Value
Mass	~80 (g)
Wing Area	108 (cm ²)
Wing Span	36 (cm)
Wing Chord	6 (cm)
Wing Aspect Ratio	6 (—)
Flapping Frequency	10–20 (Hz)
Flapping Domain	50 (deg)
Upstroke Angle	22.5 (deg)
Downstroke Angle	-22.5 (deg)
Body Length	12 (cm)
Propulsion System	Electric Motor
Motor Type Selection	Brushless Motor

3.2 Aerodynamics and Flight Dynamics Principles in Micro Dimensions

Flight in micro dimensions has specific characteristics due to the low Reynolds number (10^2 – 10^4) that distinguishes it from conventional aviation. Under these conditions, the flow around the wings is unstable and transient, and flapping, in addition to generating lift, also creates thrust. This feature gives Flapping-Wing Micro Aerial Vehicles (FW-MAVs) an advantage in terms of maneuverability and efficiency compared to fixed-wing and propeller structures.

The main aerodynamic mechanisms at this scale include the formation of the delayed stall vortex, Wake Capture, and Clap-and-Fling phenomena, all of which contribute to increasing force. Also, wing flexibility plays an important role in regulating the angle of attack and improving efficiency. Experimental and biological studies on insects and small birds have shown that the wing tip path (such as an elliptical path or figure-eight) and large flapping amplitude (90 to 140 degrees) are very effective in improving flight stability and efficiency.

Various methods are used to model this flight. Quasi-Steady models are used for rapid force estimation, while CFD simulations offer the highest accuracy but are very time-consuming due to complexity and the need for powerful hardware. In addition, analytical methods based on energy theory and Slipstream have been developed to estimate power consumption and flight performance.

In the present project, design and analysis were performed using SolidWorks for 3D modeling and MATLAB and Python codes for dynamic and kinematic analysis of mechanisms. Past studies and experiences, from early prototypes like Microbat to advanced designs like Dove and BionicOpter, all show that combining lightweight and optimal mechanical design with precise aerodynamic analysis is the key to achieving stable and efficient flight in flapping-wing MAVs.

3.2.1 Force Modeling on the Wing

The total aerodynamic forces on a flapping wing are significantly influenced by its deformation during flight. As illustrated by [1], specifically in their Figure 10, both spanwise bending and twisting of the wing contribute distinct force components in the sensor frame.

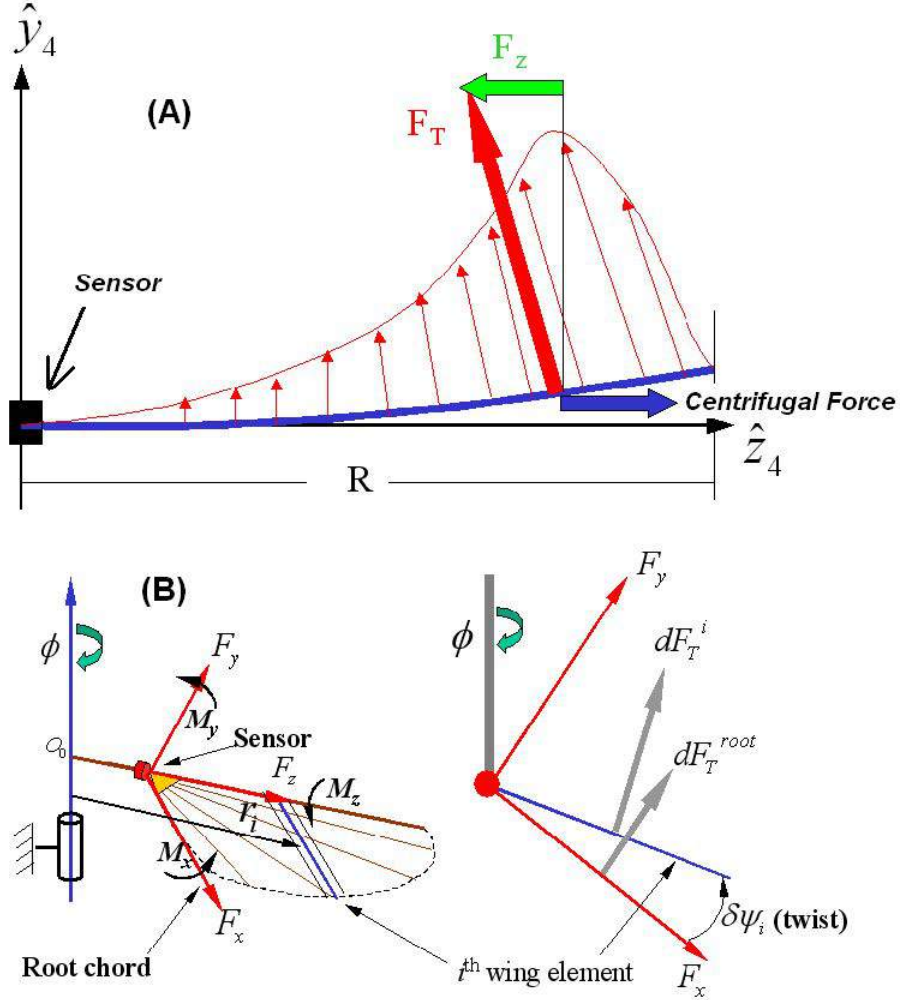


Figure 3.1: (A) Shows how the spanwise bending creates a $-F_z$ component of the aerodynamic force which acts in $-z_4$ direction in the sensor frame. The centrifugal force acts in the sensor $+z_4$ direction. (B) Shows how the spanwise twist $\delta\psi_i$ at the i^{th} element contributes to the $-F_x$ component of aerodynamic force in the sensor frame [1].

Wing Deformation Effects on Aerodynamic Forces

Wing deformations during flapping flight induce specific components of aerodynamic force in the sensor frame:

1. **Spanwise Bending:** As shown in Figure 3.1(A), spanwise bending of the wing creates a component of the aerodynamic force acting in the $-F_z$ direction (per-

pendicular to the stroke plane) of the sensor frame. This bending-induced force often counteracts the centrifugal force.

2. **Spanwise Torsion:** Figure 3.1(B) illustrates that spanwise twist ($\delta\psi_i$) along the wing's elements contributes to an aerodynamic force component in the $-F_x$ direction (parallel to the leading edge) of the sensor frame. This twisting also influences the effective angle of attack and the overall force distribution.

Centrifugal Force

In addition to aerodynamic forces, mechanical forces such as centrifugal force play a role. As depicted in Figure 3.1(A), due to the rotation of the wing around the base axis, a centrifugal force is applied tangentially outwards along the wing, acting in the $+z_4$ direction of the sensor frame. This force is a product of the wing's mass and its rotational acceleration.

Sensor Frame

A multi-axis force-torque sensor is typically installed at the wing base. This sensor measures forces (F_x, F_y, F_z) and torques (M_x, M_y, M_z) in its local coordinate system, allowing these raw measurements to be converted into aerodynamic components for detailed analysis.

Resultant Force Angle

The angle of the resultant total aerodynamic force (F_T) is a complex function of the lift force distribution along the wing and its instantaneous twist. This angle is determined by summing the vector components of the forces acting on each element of the wing.

3.2.2 Lift Distribution on the Wing

The MATLAB code generated based on the force model discussed in Section 3.2.1 calculates the lift force distribution along the wing based on aerodynamic principles. The process is as follows:

Wing Span Discretization

The length of the semi-wing is divided into small elements, and the Lift force is calculated at each longitudinal position (x).

Aerodynamic Force Calculations

The lift force per unit length $L(x)$ is calculated using the following relationship:

$$L(x) = \frac{1}{2} \rho V(x)^2 C_L(x) c(x) \quad (3.1)$$

3.2. AERODYNAMICS AND FLIGHT DYNAMICS PRINCIPLES IN MICRO DIMENSIONS

The meanings and values of the symbols inside the formula are presented in the table below:

Table 3.3: Meanings and Values of Lift Force Distribution Formula Coefficients

Parameter	Description / Value
Assumptions	$C_{L,max} = 1.2$, Altitude = 1200 m, flapping frequency = 10 Hz.
ρ	Air density (considered for Tehran).
$V(x)$	Local velocity at position (x), which depends on oscillation frequency, angular velocity, and wing kinematics.
$C_L(x)$	Lift coefficient modeled elliptically: $C_L(x) = C_{L,max} \sqrt{1 - (x/r)^2}$
$c(x)$	Chord length at position (x).

Considering the presented model for lift distribution on a wing and the input parameter values of the ornithopter given in Table 3.2, we obtain the following results.

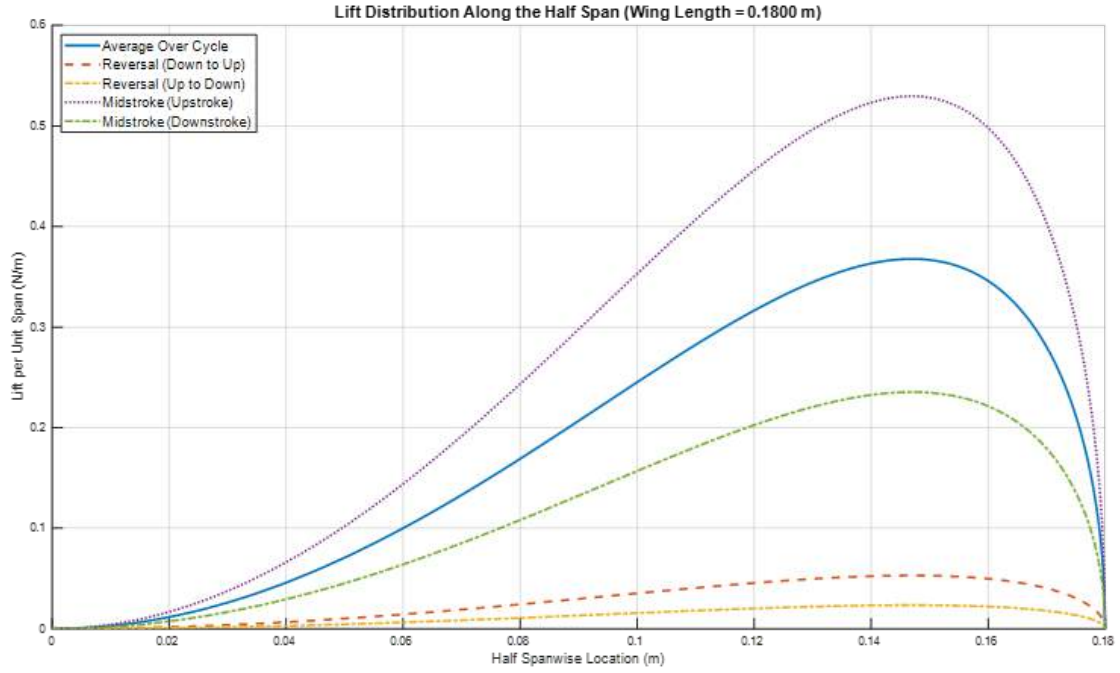


Figure 3.2: Lift force distribution on the wing

Different Phases of Lift Force: Different lift states (average, mid-stroke, and return) are calculated by changing velocity characteristics:

- **Mid-Upstroke/Downstroke:** Maximum lift force is generated in the middle of the stroke due to high angular velocity.
- **Return:** Lift force decreases, which is due to the reduction in angular velocity during transition stages.

Integration for Total Lift Force Calculation:

$$L_{total} = \int_0^r L(x) dx \quad (3.2)$$

The total lift force for both front and rear wings is calculated by multiplying the total lift force of one wing by 4.

Visualization

The output chart in Figure 3.2 shows how the lift force changes along the wing for different stages of the oscillation cycle:

- **Mid-Upstroke Distribution:** Has the highest peak due to high angular velocity and effective angle of attack.
- **Average Lift Force:** Presents a smooth view of the entire cycle, used to calculate the total lift force of one wing from this distribution.

3.2.3 Relation Between Force Modeling and Lift Distribution

Force Modeling (Figure 3.1): Integrates aerodynamic forces (F_T) applied perpendicular to the chord along the wing and considers the effects of wing bending and twisting. It is calculated based on experimental data and quasi-steady/unsteady aerodynamic models.

Numerical Lift Distribution (Figure 3.2): Numerical simulation of lift forces along the wing focuses on aerodynamic principles and does not consider the effects of structural deformation.

These two analyses complement each other; force modeling confirms experimental observations, while numerical lift distribution provides insight into aerodynamic force generation and yields an estimate for the lift generated by each wing, which is vital for sizing the ornithopter.

3.2.4 Extraction of Aerodynamic Characteristics Using Software

The software commonly used in various aerospace tasks for modeling and simulating aircraft aerodynamics with various VLM and UVLM solutions is OpenVSP. However, a significant limitation of this platform is that it primarily supports fixed-wing aircraft modeling. For flapping-wing vehicles, such as the one under investigation, direct modeling and simulation within OpenVSP are not possible. Therefore, Ptera was selected as the primary software tool, and the reasons for this choice over OpenVSP are detailed below:

The Ptera software package is specifically designed for the dynamic analysis and airflow simulation around aircraft structures with oscillating wings [15]. This tool offers distinct advantages over OpenVSP for aerodynamic simulation of flapping-wing vehicles:

1. **Suitability for Flapping Wing Aerodynamics:** Ptera is specifically developed for analyzing structures with complex flapping motions, including dynamic wing deformation, oscillation, and changes in angle of attack. It accurately models fluid-structure interaction, phenomena like Dynamic Stall, and is adept at simulating

nonlinear and unsteady flows, which are critical in flapping flight. OpenVSP, in contrast, is primarily geared towards steady-flow aerodynamics and fixed-wing design, lacking the specialized capabilities for dynamic flapping motions and their associated complex fluid phenomena.

2. **Bio-inspired and Flexible Structure Analysis:** Ptera offers robust tools for simulating bio-inspired flight behavior and analyzing flexible structures. This includes capabilities for setting up and testing flapping motion control, which is essential for designing flexible structures and evaluating their energy efficiency in bio-inspired aircraft. OpenVSP, being more of a geometric-analytical tool, is less suited for these advanced dynamic and bio-inspired applications.
3. **Precision in Flapping Flight Data:** Ptera provides precise data on force distribution, torques, and energy required for oscillating motions, which is vital for evaluating and optimizing flapping-wing vehicle designs. While OpenVSP can perform aerodynamic analyses, it may not offer sufficient detail for the advanced and specific requirements of flapping flight.

Typically, such simulations are complemented by laboratory and experimental tests to extract and validate aerodynamic parameters. Since our intended aircraft had not yet reached the physical construction stage, and thus a physical prototype was unavailable, two main solutions were considered for model validation. The first involved using software specifically developed for flapping-wing aerodynamic analysis. Initially, a Python-based package named Ptera, published approximately few years ago and available on GitHub, was explored for its steady and unsteady simulation capabilities for flapping-wing vehicles. An attempt was made to perform simulations using available FWMAV information (dimensions, geometry, flapping frequency, etc.) and estimating missing parameters (e.g., airfoil type). However, these simulations yielded unreasonable results. After thorough review, it became apparent that the software package did not fully support our specific four-wing configuration with identical cycles, and would have required extensive coding from scratch, which was not feasible within the project's time constraints.

Given these computational limitations, the second solution was adopted, which involved utilizing empirical data and existing statistics from similar flapping-wing vehicles, such as the BionicOpter, to inform our design. Based on this approach and findings from various studies on similar FWMAVs, the lift coefficient of the FWMAV in a full flapping cycle is generally expected to fall within the range of -1.5 to $+1.5$. These empirical values were then used to ensure consistency with the lift coefficients employed in our calculations.

3.3 Mechanism Design Criteria

In designing the mechanism of flapping-wing vehicles, several key principles are proposed:

- **Weight and Simplicity:** Every extra gram has a negative effect on flight performance, so the mechanism must be lightweight and have few parts [8, 9].

- **Symmetry and Stability:** Wings must have in-phase and symmetric motion to prevent deviation and instability [9].
- **Suitable Amplitude and Frequency:** The mechanism must be able to create a large flapping angle (about 40–140 degrees) with high frequency (10–100 Hz) [5].
- **Controllability:** The design must allow for changing amplitude, phase, or angle of attack to provide the possibility of applying flight controls [4, 16].

3.4 Design and Simulation Tools

To achieve these criteria, a set of tools is used:

- **3D Modeling (SolidWorks):** Drawing and examining mechanisms and evaluating spatial limitations.
- **Dynamic and Kinematic Analysis (Python/MATLAB):** Simulation of linkage motion, calculation of angle, velocity, and acceleration of wings, and investigation of dynamic forces.
- **Aerodynamic Analysis:** Use of simple quasi-steady models for rapid force estimation and in special cases, utilizing CFD for precise flow analysis. Although CFD simulation offers valuable results, it requires powerful hardware and long time due to very heavy calculation volume, which was beyond the scope of this project due to computational resource constraints.

Chapter 4

Development of Mechanisms

In this chapter, the four main actuator mechanisms are examined. For each mechanism, the design principles, governing equations, and in some cases, the construction of a prototype are described, followed by an analysis of their performance and limitations. The aim of this section is to assess the feasibility and present improvement paths for employing these mechanisms in micro-scale flapping-wing aerial systems.

4.1 Coil and Magnet Mechanism

Mechanisms based on electromagnetic actuators hold a special place in research and design of micro aerial vehicles due to their simplicity of construction, compact dimensions, and the possibility of direct current controllability. The basis of this mechanism is the interaction between the magnetic field produced by a current-carrying coil and the magnetic moment of a permanent magnet. When electric current passes through the coil, a magnetic field is generated; the interaction of this field with the permanent magnet produces a torque that causes angular motion. Reversing the current direction reverses the magnetic field polarity and, consequently, the direction of the torque. In this manner, a reciprocating oscillatory motion is created, which can serve as the primary actuator for flapping-wing propulsion when properly designed.

4.1.1 Design and Performance

This section presents the design parameters, governing relationships, and equations of motion for the coil-magnet mechanism in detail.

Coil Specifications

The electromagnetic coil was custom-wound with the following measured specifications:

- Wire diameter: 0.05 mm (copper, enameled)
- Number of turns (N): 1155

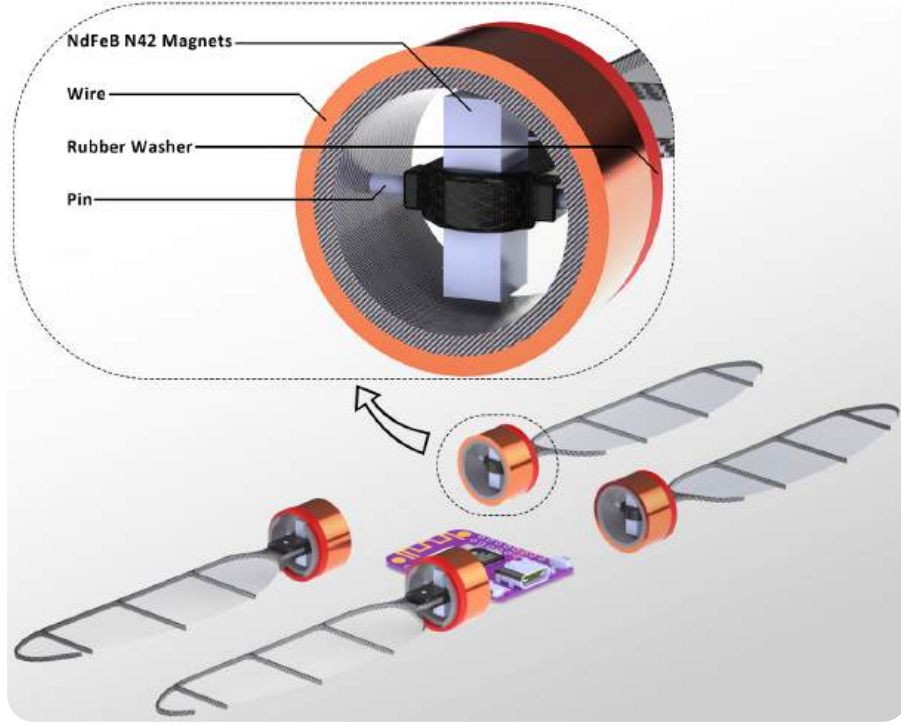


Figure 4.1: CAD model of the coil and magnet mechanism

- Total wire length: 40–50 m
- Coil geometry (cylindrical):
 - Axial length (L_c): 10 mm
 - Inner diameter (D_{in}): 12 mm
 - Outer diameter (D_{out}): 16 mm (including winding and adhesive)
- Measured resistance (R_{wire}): 320 Ω
- Operating frequency range: up to 20 Hz

Parameters Table

Table 4.1 summarizes the calculated and measured parameters used in the design and analysis of the coil-magnet mechanism. The derivation of each parameter is presented in the following subsections.

Theoretical Basis

The electromagnetic torque exerted on a magnetic dipole placed in an external magnetic field depends on the dipole moment, the field strength, and the relative orientation between them:

$$\tau_{em}(t) = m_{mag} \cdot B_{coil}(t) \cdot \sin(\theta(t)) \quad (4.1)$$

where:

4.1. COIL AND MAGNET MECHANISM

Table 4.1: Key parameters of the coil-magnet mechanism

Symbol	Description	Value
Coil Parameters		
L_c	Axial coil length	0.01 m
D_{in}	Inner diameter	0.012 m
D_{out}	Outer diameter	0.016 m
R	Mean coil radius	0.007 m
N	Number of turns	1155
R_{wire}	Measured coil resistance	320 Ω
L_{coil}	Coil inductance	16 mH
τ_{RL}	RL time constant	0.05 ms
k_B	Magnetic field coefficient	0.084 T/A
Magnet Parameters		
B_r	Residual magnetic flux density (N42 NdFeB)	1.3 T
V_m	Magnet volume ($10 \times 3 \times 2$ mm)	6×10^{-8} m ³
m_{mag}	Magnetic dipole moment	0.062 A·m ²
M	Magnet mass	9×10^{-4} kg
Derived Parameters		
k_τ	Torque coefficient	5.2×10^{-3} N·m/A
d	Distance from pivot to magnet center of mass	~5 mm
γ	Viscous friction coefficient	Negligible

- m_{mag} : magnetic dipole moment of the permanent magnet (A·m²)
- $B_{coil}(t)$: magnetic field produced by the coil at the magnet location (T)
- $\theta(t)$: angular displacement of the magnet relative to the field axis (rad)

This relationship forms the foundation of the dynamic analysis and mechanism sizing.

Magnetic Field Calculation

For a finite-length solenoid with axial length L_c , mean radius R , and N turns, the on-axis magnetic field at the coil center is given by:

$$B_{coil} = \mu_0 \frac{N}{L_c} I(t) \cdot \underbrace{\frac{L_c}{\sqrt{L_c^2 + 4R^2}}}_{\text{finite-length correction}} \quad (4.2)$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space.

Substituting the coil parameters ($L_c = 0.01$ m, $R = 0.007$ m, $N = 1155$):

$$\text{Correction factor} = \frac{L_c}{\sqrt{L_c^2 + 4R^2}} = \frac{0.01}{\sqrt{0.0001 + 0.000196}} = \frac{0.01}{0.0172} \approx 0.58 \quad (4.3)$$

4.1. COIL AND MAGNET MECHANISM

This indicates that the coil produces approximately 58% of the field that an ideal infinite solenoid with the same turn density would generate.

The magnetic field can therefore be expressed as:

$$B_{coil} = (4\pi \times 10^{-7}) \times \frac{1155}{0.01} \times 0.58 \times I(t) \approx 0.084 I(t) \text{ [T]} \quad (4.4)$$

Thus, the magnetic field coefficient is $k_B = 0.084 \text{ T/A}$.

Coil Inductance

The inductance of a single-layer solenoid with finite length is approximated by the Nagaoka formula:

$$L_{coil} = \mu_0 \frac{N^2 \pi R^2}{L_c + 0.9R} \quad (4.5)$$

Substituting the measured parameters:

$$L_{coil} = \frac{(4\pi \times 10^{-7}) \times (1155)^2 \times \pi \times (0.007)^2}{0.01 + 0.9 \times 0.007} = \frac{2.58 \times 10^{-4}}{0.0163} \approx 0.016 \text{ H} = 16 \text{ mH} \quad (4.6)$$

The RL time constant of the coil circuit determines how quickly the current can respond to voltage changes:

$$\tau_{RL} = \frac{L_{coil}}{R_{wire}} = \frac{0.016}{320} = 5 \times 10^{-5} \text{ s} = 0.05 \text{ ms} \quad (4.7)$$

Since the intended operating frequency is up to 20 Hz (period = 50 ms), the time constant of 0.05 ms is three orders of magnitude smaller. This confirms that the electrical dynamics are sufficiently fast and will not limit the mechanical response.

Magnet Properties

The selected permanent magnet is a neodymium-iron-boron (NdFeB) type, grade N42, with dimensions $10 \times 3 \times 2 \text{ mm}$. The magnetic dipole moment of a uniformly magnetized body is:

$$m_{mag} = \frac{B_r}{\mu_0} \cdot V_m \quad (4.8)$$

where B_r is the residual flux density and V_m is the magnet volume.

The magnet volume is:

$$V_m = 10 \times 3 \times 2 \text{ mm}^3 = 60 \text{ mm}^3 = 6 \times 10^{-8} \text{ m}^3 \quad (4.9)$$

Substituting $B_r = 1.3 \text{ T}$ for N42-grade NdFeB:

$$m_{mag} = \frac{1.3}{4\pi \times 10^{-7}} \times 6 \times 10^{-8} = \frac{7.8 \times 10^{-8}}{1.257 \times 10^{-6}} \approx 0.062 \text{ A}\cdot\text{m}^2 \quad (4.10)$$

Torque Coefficient

Combining the magnetic dipole moment with the field coefficient, the electromagnetic torque can be written as:

$$\tau_{em}(t) = m_{mag} \cdot B_{coil}(t) \cdot \sin \theta = k_{\tau} \cdot I(t) \cdot \sin \theta \quad (4.11)$$

where the torque coefficient is:

$$k_{\tau} = m_{mag} \times k_B = 0.062 \times 0.084 = 5.2 \times 10^{-3} \text{ N}\cdot\text{m/A} \quad (4.12)$$

Equation of Motion

The rotational dynamics of the magnet about its pivot are governed by Newton's second law for rotation:

$$I_m \ddot{\theta} = \sum \tau = \tau_{em} + \tau_{grav} + \tau_{fric} \quad (4.13)$$

where I_m is the moment of inertia of the magnet and attached components about the pivot axis.

The individual torque contributions are:

- **Electromagnetic torque:** $\tau_{em} = k_{\tau} I(t) \sin \theta$
- **Gravitational torque:** $\tau_{grav} = -Mgd \cos \theta$ (restoring torque due to offset center of mass)
- **Frictional torque:** $\tau_{fric} = -\gamma \dot{\theta}$ (viscous damping, assumed small)

Substituting these expressions:

$$I_m \ddot{\theta} = k_{\tau} I(t) \sin \theta - Mgd \cos \theta - \gamma \dot{\theta} \quad (4.14)$$

Assuming friction is negligible ($\gamma \approx 0$), and using the numerical values from Table 4.1:

$$I_m \ddot{\theta} = 5.2 \times 10^{-3} \cdot I(t) \sin \theta - (9 \times 10^{-4})(9.81)(5 \times 10^{-3}) \cos \theta \quad (4.15)$$

$$I_m \ddot{\theta} = 5.2 \times 10^{-3} \cdot I(t) \sin \theta - 4.4 \times 10^{-5} \cos \theta \quad (4.16)$$

4.1.2 Results and Limitations

Current Limitations

A critical limitation of this mechanism is the achievable driving current. With a coil resistance of 320 Ω , the maximum steady-state current is constrained by the supply voltage:

- At $V = 3.7 \text{ V}$ (single LiPo cell): $I_{max} = 3.7/320 = 11.6 \text{ mA}$
- At $V = 7.4 \text{ V}$ (2S LiPo): $I_{max} = 7.4/320 = 23.1 \text{ mA}$
- At $V = 12 \text{ V}$ (3S LiPo): $I_{max} = 12/320 = 37.5 \text{ mA}$

Maximum Achievable Torque

The peak electromagnetic torque (at $\theta = 90^\circ$ where $\sin \theta = 1$) is:

$$\tau_{em,max} = k_T \cdot I_{max} \quad (4.17)$$

- At $I = 11.6$ mA: $\tau_{em,max} = 5.2 \times 10^{-3} \times 0.0116 = 6.0 \times 10^{-5}$ N·m
- At $I = 37.5$ mA: $\tau_{em,max} = 5.2 \times 10^{-3} \times 0.0375 = 2.0 \times 10^{-4}$ N·m

For comparison, the gravitational torque on the magnet is approximately 4.4×10^{-5} N·m (from Equation 4.16). This means the electromagnetic torque is of the same order of magnitude as the gravitational load on the magnet alone—before any wing loads are added.

Simulation Results

Numerical simulation of Equation 4.14 was performed in Python to evaluate the dynamic response. The results, shown in Figure 4.2, confirm that:

- The electrical response is very fast (time constant $\ll$ mechanical period), so current tracks the input voltage well.
- The generated electromagnetic torque remains in the range of 10^{-5} to 10^{-4} N·m.
- This torque is comparable to resisting torques from gravity alone and is insufficient to drive any meaningful wing load.
- Angular displacement becomes severely limited, and the motion amplitude required for effective flapping (typically 30 – 60°) cannot be achieved.

Experimental Validation

To validate the theoretical analysis and simulation predictions, a physical prototype was constructed according to the specifications in Table 4.1. The fabricated sample is shown in Figure 4.3.

Experiments performed on the physical prototype confirmed the simulation results:

- Small oscillation amplitudes were observed even without attached wing loads.
- Adding even minimal external mass further reduced the motion amplitude.
- At the intended operating frequencies (10 – 20 Hz), the mechanism could not sustain oscillation against mechanical loads.

4.1. COIL AND MAGNET MECHANISM

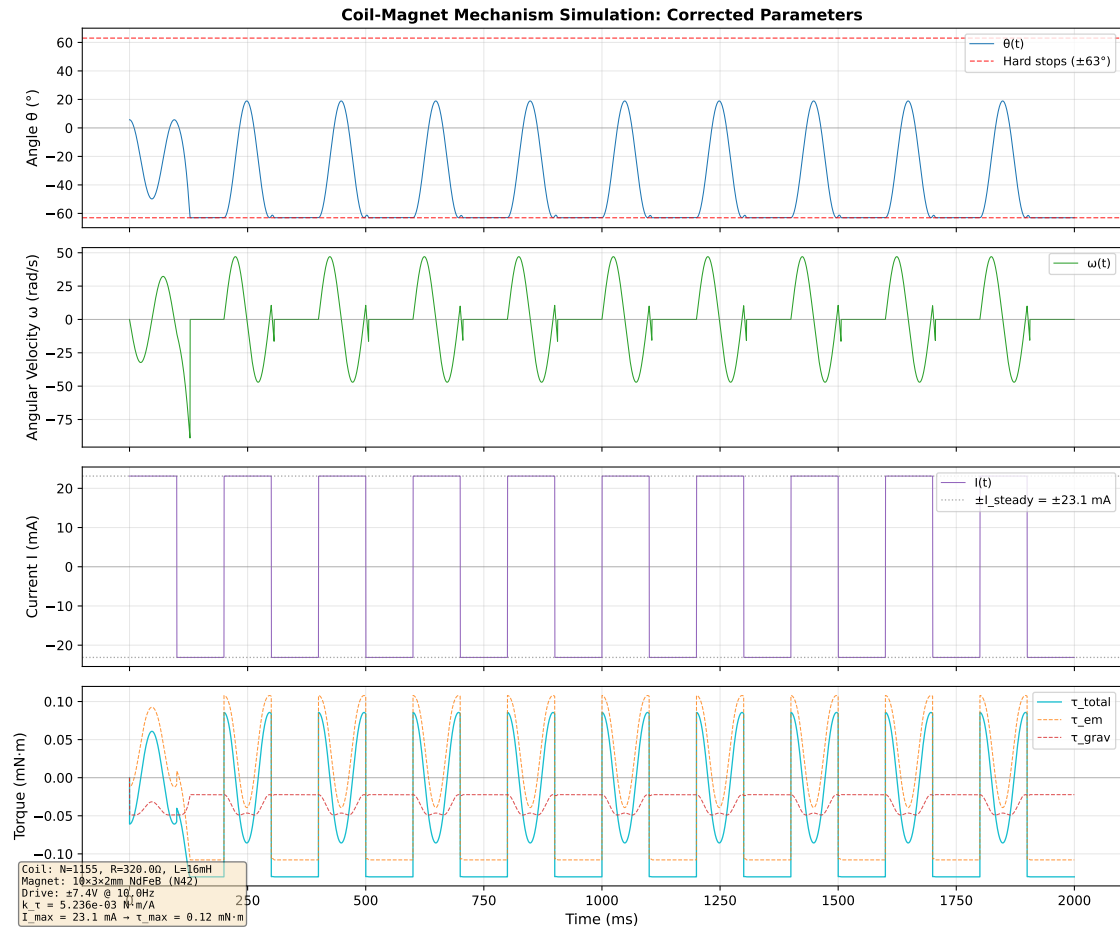


Figure 4.2: Simulation results of the coil-magnet mechanism: angular displacement, angular velocity, driving current, and electromagnetic torque versus time. The limited motion amplitude under realistic loads is evident.

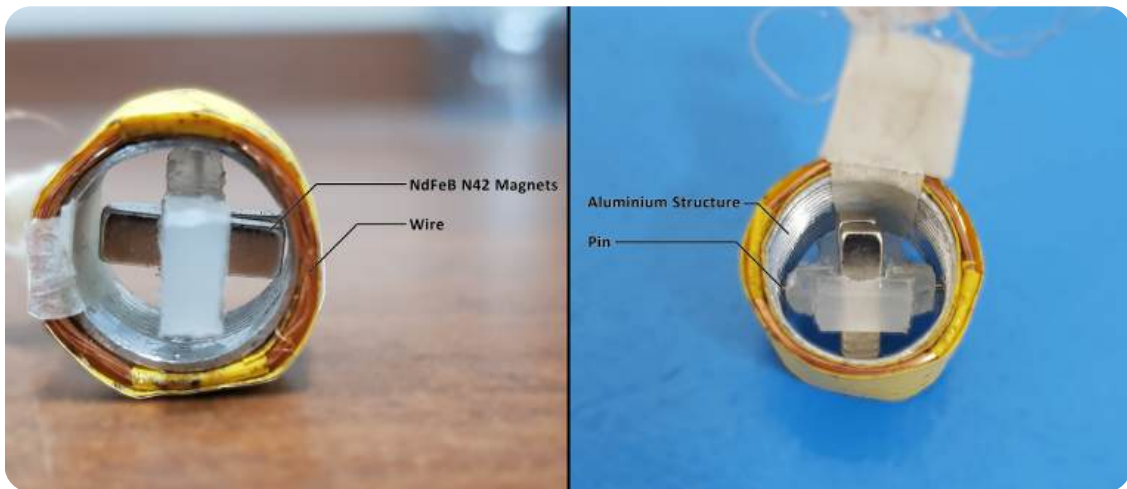


Figure 4.3: Fabricated prototype of the coil and magnet mechanism

Conclusion

The coil-magnet mechanism, in its current configuration, produces insufficient torque for practical flapping-wing actuation. The fundamental limitation is the high coil resistance ($320\ \Omega$), which restricts the driving current to tens of milliamps and limits the electromagnetic torque to the 10^{-5} – 10^{-4} N·m range. This is inadequate for overcoming inertial and aerodynamic loads associated with wing motion. Consequently, this mechanism was deemed unsuitable for further development in its present form, and attention was directed to alternative actuation approaches.

4.1.3 Potential Improvements for Future Work

Although the coil-magnet mechanism proved impractical for the current application, the following improvements could enhance its performance for future electromagnetic actuator designs:

Reduce Coil Resistance The primary limitation is the high resistance arising from the use of very fine wire (0.05 mm diameter) over a long length (40–50 m). Using thicker wire would reduce resistance significantly:

- Doubling the wire diameter to 0.1 mm reduces resistance by a factor of 4.
- This would allow $4\times$ higher current for the same voltage, yielding $4\times$ higher torque.
- Trade-off: Increased wire diameter means fewer turns for the same coil volume, partially offsetting the gain.

An optimized design would balance wire gauge, turn count, and coil geometry to maximize torque-per-watt [9].

Use Stronger Magnets or Larger Magnet Volume Increasing the magnetic dipole moment m_{mag} directly increases the torque coefficient k_τ . Options include:

- Using higher-grade NdFeB magnets (e.g., N52 with $B_r \approx 1.45$ T vs. N42's 1.3 T).
- Increasing magnet volume, though this adds mass and inertia.

Reduce Eddy Current Losses If a metallic coil former (e.g., aluminum tube) is used, eddy currents will be induced, causing energy dissipation and producing a damping force that opposes motion. This can be mitigated by:

- Using non-conductive formers (plastic, composite).
- Adding longitudinal slots to metallic formers to interrupt eddy current paths.

Studies on electromagnetic actuators for flapping-wing vehicles have shown that material selection significantly affects efficiency [3, 9].

Improve Power Electronics Higher driving voltages would proportionally increase current (and thus torque) for the same coil. Alternatively, resonant driving circuits can be used to reduce reactive power losses and enhance efficiency at the target oscillation frequency [7].

Hybrid Power Sources Supercapacitors can provide high instantaneous current that batteries cannot sustain. A hybrid system combining supercapacitors for peak current delivery with batteries for energy storage has proven effective in micro-vehicle applications such as Microbat [8]. Future designs might also incorporate solar cells for extended endurance [6].

Miniaturization via MEMS At smaller scales, electromagnetic actuators can achieve higher force-to-weight ratios due to favorable scaling of magnetic forces. MEMS fabrication techniques enable the creation of integrated coil-magnet systems with reduced part count and minimized mechanical losses [10].

In summary, while the current coil-magnet design is not viable, electromagnetic actuation remains a promising approach if the resistance-current limitation can be addressed through optimized coil design, improved materials, and advanced power electronics. These directions align with ongoing research in electromagnetic micro-actuators for flapping-wing vehicles [2, 3].

4.2 Coreless Motor with Crank-Slider Mechanism

This mechanism employs four independent DC coreless motors, each driving a single-stage gear reduction and crank-slider linkage to convert continuous rotation into reciprocating linear motion. Each motor-gear-crank assembly controls one wing, allowing independent adjustment of flapping frequency for each of the four wings. The linear displacement of each slider actuates its corresponding wing by pushing on a pivot point near the wing root, causing the wing to oscillate about its hinge.

- **Advantages:** Compact dimensions (7 mm diameter motors), low total motor mass ($4 \times 2.5 \text{ g} = 10 \text{ g}$), independent frequency control for each wing enabling differential thrust and maneuverability, adjustable stroke amplitude through crank radius selection.
- **Challenges:** Synchronization of four motors, gear mesh losses, slider friction, torque pulsation at crank dead points, and increased wiring complexity.

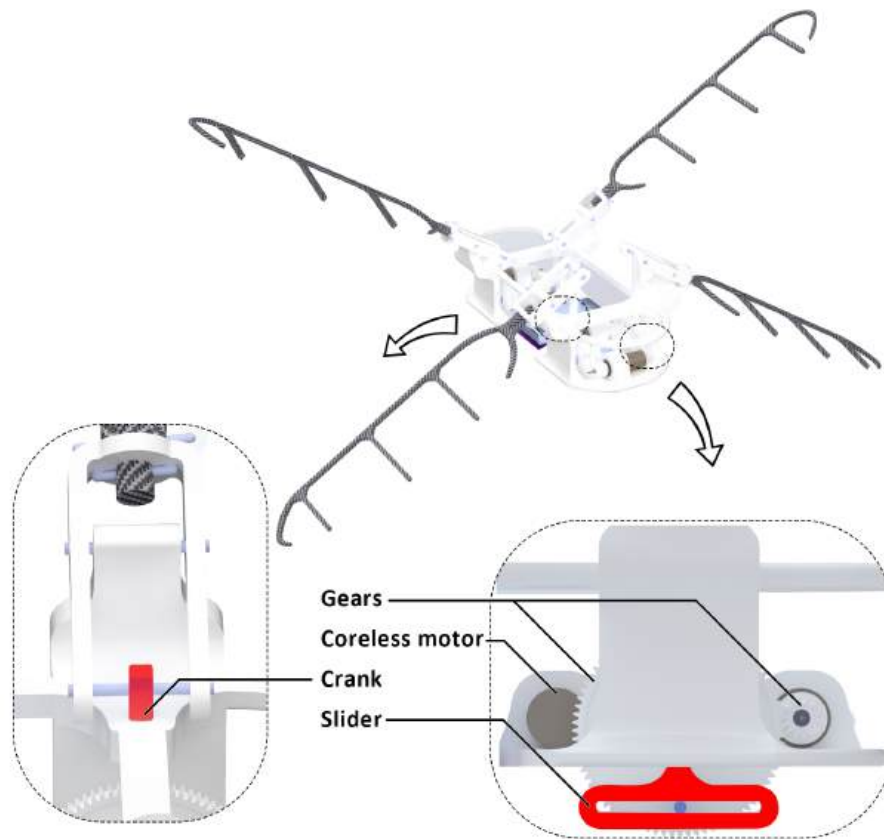


Figure 4.4: CAD model of the coreless motor mechanism with single-stage gear reduction and crank-slider linkage. Four identical assemblies are used, one for each wing.

4.2.1 Design and Performance

Each wing is driven by an identical motor-gear-crank assembly. A small pinion gear (14 teeth) mounted on the motor shaft meshes with a larger gear (79 teeth). The large gear carries an eccentric pin at radius $R = 8.7$ mm from its center, which acts as the crank. This pin slides through a vertical slider that transmits the motion to the wing root.

Motor Specifications

Each of the four coreless DC motors has the following specifications:

- Diameter: 7 mm, Length: 16.5 mm
- Shaft: $\varnothing 0.8$ mm $\times$ 6.3 mm
- Mass: 2.5 g (total for 4 motors: 10 g)
- Rated voltage: 3.7 V
- Rated current: 100 mA
- No-load speed: 50,000 rpm

From these specifications, the motor electrical parameters are derived:

$$\omega_{no-load} = 50000 \times \frac{2\pi}{60} = 5236 \text{ rad/s} \quad (4.17)$$

$$k_e = k_t = \frac{V}{\omega_{no-load}} = \frac{3.7}{5236} = 7.07 \times 10^{-4} \text{ V}\cdot\text{s/rad} \quad (4.18)$$

The terminal resistance is estimated from typical stall current characteristics:

$$R_m \approx \frac{V}{I_{stall}} \approx \frac{3.7}{0.5 \text{ to } 1.0} \approx 4\text{--}7 \ \Omega \quad (4.19)$$

A value of $R_m = 5 \ \Omega$ is used for subsequent analysis.

Gear Train Configuration

The power transmission uses a single-stage gear reduction:

- **Pinion (motor shaft):** 14 teeth
- **Driven gear (crank carrier):** 79 teeth
- **Gear ratio:** $G = z_{out}/z_{in} = 79/14 = 5.64$
- **Crank pin radius:** $R = 8.7$ mm from gear center

The crank pin on the large gear slides through a vertical slot in the slider mechanism, converting the gear's rotation into reciprocating vertical motion with stroke $2R = 17.4$ mm.

Parameters Summary

Tables 4.2 and 4.3 summarize the key parameters organized by subsystem.

Table 4.2: Motor and gear parameters (per wing, $\times 4$ total)

Parameter	Symbol	Value	Unit
Motor Parameters (each of 4 motors)			
Supply voltage	V	3.7	V
Terminal resistance	R_m	5	Ω
Torque constant	k_t	7.07×10^{-4}	N·m/A
Back-EMF constant	k_e	7.07×10^{-4}	V·s/rad
Viscous friction	b_m	≈ 0	N·m·s/rad
No-load speed	ω_{nl}	5236	rad/s
Motor mass	m_{motor}	2.5	g
Single-Stage Gear Reduction			
Pinion teeth (on motor)	z_{in}	14	—
Gear teeth (crank carrier)	z_{out}	79	—
Gear ratio	$G = z_{out}/z_{in}$	5.64	—
Mesh efficiency	η_g	0.96	—
Crank-Slider			
Crank radius (pin offset)	R	8.7	mm
Slider stroke	$2R$	17.4	mm
Slider mass	m_s	3	g
Slider viscous friction	b_s	0	N·s/m
Slider Coulomb friction	F_μ	0	N

Kinematic Relations

The kinematic chain from motor to wing is analyzed for each motor-wing assembly:

Motor to Gear Output The motor angular velocity is reduced by the single-stage gear ratio G :

$$\omega_{out} = \frac{\omega_m}{G}, \quad \theta_{out} = \frac{\theta_m}{G} \quad (4.20)$$

where subscript “out” denotes the output gear (crank carrier) and “m” denotes the motor shaft.

For a flapping frequency of $f_{out} = 7$ Hz:

$$\omega_{out} = 2\pi f_{out} = 2\pi \times 7 = 44.0 \text{ rad/s} \quad (4.21)$$

$$\omega_m = G \times \omega_{out} = 5.64 \times 44.0 = 248 \text{ rad/s} = 2370 \text{ rpm} \quad (4.22)$$

This motor speed is only 4.7% of the no-load speed (50,000 rpm), indicating the motor operates at very low speed with substantial torque capability available.

4.2. CORELESS MOTOR WITH CRANK-SLIDER MECHANISM

Table 4.3: Wing geometry and dynamics parameters (per wing)

Parameter	Symbol	Value	Unit
Wing Geometry			
Hinge-to-actuation distance	a	18	mm
HA line orientation	α	0	rad
Maximum wing angle	$\phi_{max} = \arcsin(R/a)$	± 28.9	deg
Wing Dynamics			
Wing inertia (about hinge)	I_w	7×10^{-6}	$\text{kg}\cdot\text{m}^2$
Hinge stiffness	k_h	0	$\text{N}\cdot\text{m}/\text{rad}$
Hinge damping	c_h	0	$\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$
Wing Loading			
Wing mass	m_w	2	g
COM distance from hinge	r_G	40	mm
COM angle offset	β_G	0	rad
Aero drag coefficient	K_d	6.5×10^{-7}	$\text{N}\cdot\text{m}/(\text{rad}/\text{s})^2$
Operating Condition			
Target flapping frequency	f_{out}	7	Hz

Crank to Slider For a crank of radius R rotating at angular velocity ω_{out} , the slider vertical position is:

$$y = R \sin(\theta_{out}) \quad (4.23)$$

Taking time derivatives (assuming constant ω_{out}):

$$\dot{y} = R \omega_{out} \cos(\theta_{out}) \quad (4.24)$$

$$\ddot{y} = -R \omega_{out}^2 \sin(\theta_{out}) \quad (4.25)$$

With $R = 8.7$ mm and $\omega_{out} = 44$ rad/s:

- Slider amplitude: ± 8.7 mm (stroke = 17.4 mm)
- Maximum slider velocity: $R\omega_{out} = 0.0087 \times 44 = 0.38$ m/s
- Maximum slider acceleration: $R\omega_{out}^2 = 0.0087 \times 44^2 = 16.8$ m/s²

Slider to Wing Angle The geometric constraint relating slider position y to wing angle ϕ is:

$$y = a \sin(\alpha + \phi) \quad (4.26)$$

where $a = 18$ mm is the distance from hinge to actuation point, and $\alpha = 0$ is the orientation of the hinge-actuation line.

Solving for the wing angle:

$$\phi = \arcsin\left(\frac{y}{a}\right) - \alpha = \arcsin\left(\frac{y}{a}\right) \quad (4.27)$$

The maximum wing angle amplitude is:

$$\phi_{max} = \arcsin\left(\frac{R}{a}\right) = \arcsin\left(\frac{8.7}{18}\right) = 28.9^\circ \approx \pm 29^\circ \quad (4.28)$$

This provides a total flapping range of approximately 58° , which is suitable for generating lift and thrust in a dragonfly-inspired design.

Differentiating the constraint equation yields the angular velocity and acceleration:

$$\dot{\phi} = \frac{\dot{y}}{\sqrt{a^2 - y^2}} \quad (4.29)$$

$$\ddot{\phi} = \frac{\ddot{y}(a^2 - y^2) + y \dot{y}^2}{(a^2 - y^2)^{3/2}} \quad (4.30)$$

Mechanical Advantage The instantaneous mechanical advantage relates infinitesimal changes in crank angle to wing angle:

$$\frac{d\phi}{d\theta_{out}} = \frac{R \cos(\theta_{out})}{a \cos(\alpha + \phi)} \quad (4.31)$$

This ratio varies throughout the cycle:

- At $\theta_{out} = 0$ or π (crank vertical): maximum angular velocity transfer
- At $\theta_{out} = \pm\pi/2$ (crank horizontal, dead points): wing velocity momentarily zero

Dynamic Relations

Wing Hinge Torque The total torque required at each wing hinge to sustain the prescribed motion is:

$$M_w = I_w \ddot{\phi} + c_h \dot{\phi} + k_h \phi + M_{grav} + M_{aero} \quad (4.32)$$

where:

$$M_{grav} = m_w g r_G \cos(\beta_G + \phi) \quad (\text{gravitational torque}) \quad (4.33)$$

$$M_{aero} = K_d \dot{\phi} |\dot{\phi}| \quad (\text{quadratic aerodynamic drag}) \quad (4.34)$$

Slider Force from Wing Load Using the principle of virtual work ($M_w \delta\phi = F_{wing} \delta y$):

$$F_{wing} = \frac{M_w}{a \cos(\alpha + \phi)} \quad (4.35)$$

The total force on each slider includes inertia and friction:

$$F_{tot} = F_{wing} + m_s \ddot{y} + b_s \dot{y} + F_\mu \text{sgn}(\dot{y}) \quad (4.36)$$

Crank Torque The torque required at each output gear (crank) is:

$$\tau_{crank} = F_{tot} \cdot R \cos(\theta_{out}) \quad (4.37)$$

Motor Requirements Reflecting the crank torque to each motor shaft through the gear:

$$\tau_m = \frac{\tau_{crank}}{G \cdot \eta_g} \quad (4.38)$$

The motor current and voltage requirements are:

$$i_m = \frac{\tau_m + b_m \omega_m}{k_t} \quad (4.39)$$

$$V_{req} = R_m i_m + k_e \omega_m \quad (4.40)$$

For the mechanism to operate as intended, the required voltage V_{req} must not exceed the supply voltage $V = 3.7$ V for each motor.

4.2.2 Results and Limitations

Operating Point Analysis

At the target flapping frequency of $f_{out} = 7$ Hz (per wing):

- Output gear speed: $\omega_{out} = 44$ rad/s
- Motor speed: $\omega_m = 248$ rad/s = 2370 rpm (4.7% of no-load)
- Back-EMF at motor: $V_{back} = k_e \omega_m = 7.07 \times 10^{-4} \times 248 = 0.18$ V
- Voltage available for torque: $V - V_{back} = 3.7 - 0.18 = 3.52$ V

The low motor speed leaves substantial voltage headroom for torque production, which is favorable for this application.

Simulation Results

A numerical simulation was performed for a single motor-wing assembly with time step $dt = 0.5$ ms over 5 seconds. Figure 4.5 presents the results.

Key observations from the simulation:

- **Slider Displacement:** Sinusoidal motion with amplitude ± 8.7 mm, consistent with the crank radius.
- **Wing Angle:** Oscillates with amplitude $\pm 28.9^\circ$ (total range $\approx 58^\circ$), providing substantial flapping motion suitable for lift generation.

4.2. CORELESS MOTOR WITH CRANK-SLIDER MECHANISM

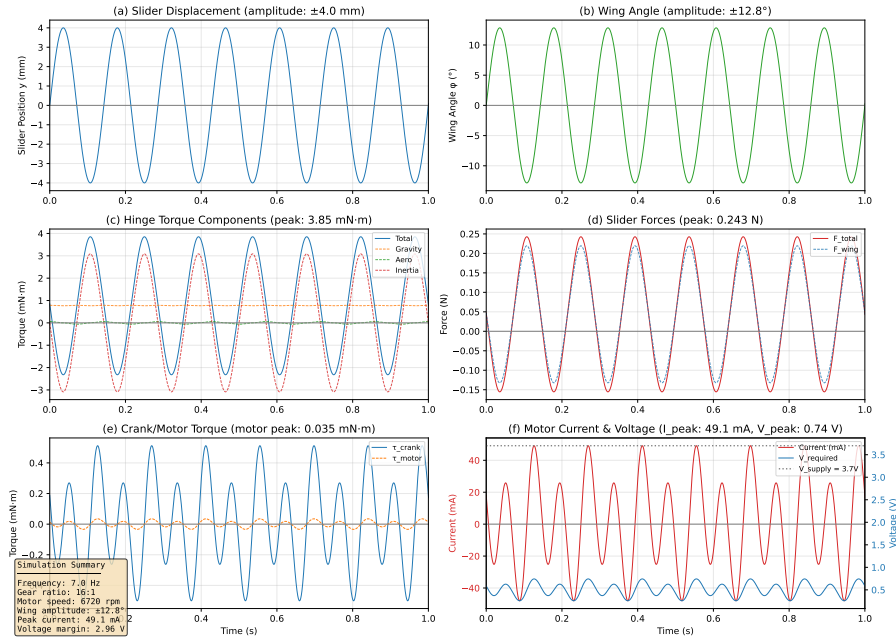


Figure 4.5: Simulation results for one coreless motor crank-slider assembly at 7 Hz flapping frequency: (a) slider displacement, (b) wing angle, (c) hinge torque components, (d) slider forces, (e) crank torque, (f) motor current and voltage.

- **Hinge Torque:** Peak values increased compared to the smaller crank design due to higher angular velocities. The gravitational component remains significant; aerodynamic drag grows with the larger amplitude motion.
- **Slider Force:** Higher than previously due to larger accelerations associated with the increased crank radius.
- **Crank Torque:** Shows characteristic double-peaked modulation per revolution, with higher peak values due to increased mechanical loads.
- **Motor Current:** Within acceptable limits for the motor's capability.
- **Required Voltage:** Remains below the 3.7 V supply, confirming feasibility at 7 Hz.

Four-Motor System Considerations

The complete mechanism employs four identical motor-gear-crank assemblies, one for each wing. This configuration offers:

- **Independent frequency control:** Each wing can operate at a different frequency, enabling differential thrust for yaw control and asymmetric lift for roll control.
- **Phase control:** The relative phase between front and rear wings can be adjusted (e.g., 90° phase difference as observed in dragonflies) to optimize aerodynamic efficiency.

- **Redundancy:** Failure of one motor affects only one wing, potentially allowing limited flight capability.
- **Total power consumption:** At 7 Hz with all four motors, the estimated current is $4 \times I_{motor} \approx 4 \times 150 \text{ mA} = 600 \text{ mA}$ from a 3.7 V supply.

However, challenges include:

- Synchronization electronics for coordinated wing motion
- Increased wiring complexity (8 wires minimum for 4 motors)
- Total motor mass of 10 g (significant for a target 80 g vehicle)

Frequency Limitations

Increasing the flapping frequency increases motor speed and torque requirements. The maximum achievable frequency is limited by voltage and current constraints:

Table 4.4: Frequency sweep analysis (per motor)

f_{out} (Hz)	Motor (rpm)	V_{req} (V)	Margin (V)	Status
5	1693	~ 1.5	2.2	✓ OK
7	2370	~ 2.0	1.7	✓ OK
10	3386	~ 2.8	0.9	✓ OK
15	5079	~ 3.5	0.2	△ Marginal
20	6771	~ 4.2	-0.5	× Exceeds

The analysis indicates that frequencies up to approximately 12–15 Hz are achievable with the 3.7 V supply, but higher frequencies would require:

- Higher supply voltage (2S LiPo at 7.4 V)
- Lower gear ratio (faster motor, less torque amplification)
- Reduced mechanical loads (lighter wings, smaller crank radius)

Prototype Validation

A physical prototype was constructed with four motor-gear-crank assemblies (Figure 4.6).

Experimental observations:

- Stable synchronized operation was achieved at frequencies up to 10 Hz
- The larger wing amplitude ($\pm 29^\circ$) produced noticeable airflow
- Motor heating was minimal at 7 Hz continuous operation
- Independent frequency variation between wings was demonstrated

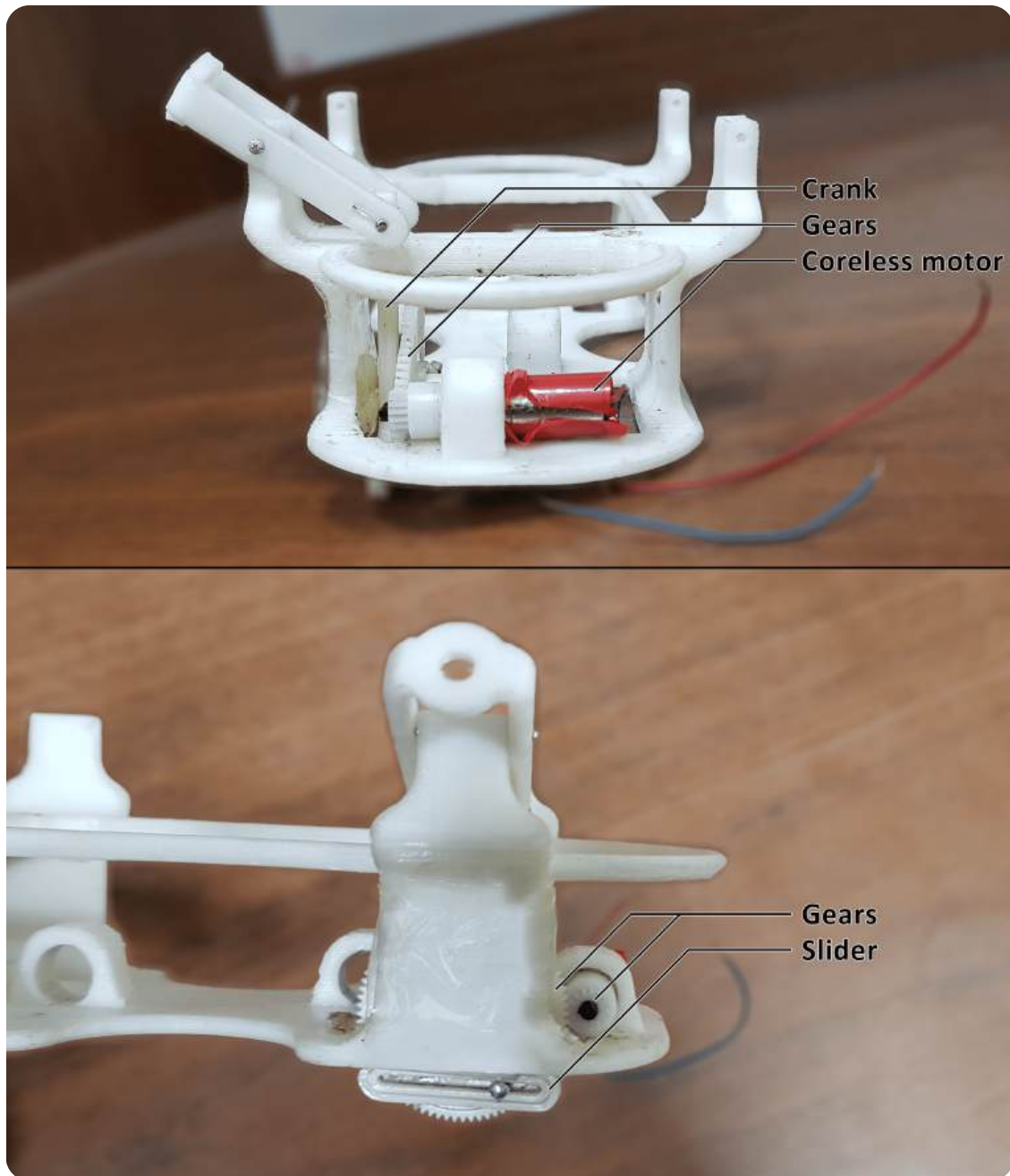


Figure 4.6: Fabricated prototype of the four-motor coreless mechanism

Conclusion

The four-motor coreless mechanism successfully demonstrates independent wing control with substantial flapping amplitude ($\pm 29^\circ$). The single-stage 14:79 gear reduction provides adequate torque amplification while keeping motor speeds low. However, the total motor mass (10 g) and wiring complexity present challenges for the target 80 g vehicle. For higher frequencies (15–20 Hz), a higher supply voltage or alternative motor selection would be required.

4.2.3 Potential Improvements

Based on the analysis, the following improvements are recommended:

Reduce Peak Torque at Dead Points Adding elastic elements (springs or compliant links) between the crank and slider can store and release energy, smoothing the motor load profile and reducing peak current draw [3, 9].

Optimize Gear Ratio The current ratio ($G = 5.64$) results in low motor speeds. A lower ratio would increase motor speed (improving efficiency) at the cost of reduced torque amplification. Systematic optimization considering motor efficiency curves could yield improved overall performance [6].

Synchronization Strategy For coordinated four-wing flight, the control system must manage:

- Phase relationships between wing pairs (e.g., front-rear phase offset)
- Frequency matching for steady flight
- Differential frequency/phase for maneuvering

A microcontroller-based PWM system with feedback could achieve this [10].

Mass Reduction The four-motor configuration contributes 10 g to the vehicle mass. Alternatives include:

- Using two motors with mechanical coupling to pairs of wings
- Smaller/lighter motors (at reduced torque capability)
- Integrated motor-gear units to reduce structural mass

[2, 5]

Higher Voltage Operation Using a 2S LiPo battery (7.4 V) would nearly double the voltage headroom, enabling:

- Higher flapping frequencies (up to 25–30 Hz potentially)
- Greater torque margin for larger wings or payloads
- Reduced sensitivity to voltage sag under load

In summary, the four-motor coreless mechanism demonstrates the feasibility of independent wing control but faces mass and complexity challenges. The substantial flapping amplitude achieved ($\pm 29^\circ$) is promising for lift generation. These findings informed the development of alternative mechanisms described in Sections 4.3 and 4.4, which explore centralized motor configurations to reduce mass and complexity [12].

4.3 Dove-Inspired Gear Mechanism

The third mechanism explored in this project draws inspiration from the Dove flapping-wing micro aerial vehicle developed at Northwestern Polytechnical University [2]. In this architecture, a single brushless motor drives both wings through a gear reduction stage, a rack-and-pinion converter, and an asymmetric four-bar linkage that transforms continuous rotation into symmetric wing oscillation.

- **Advantages:** Single motor reduces mass and complexity compared to multi-motor designs; high power transmission efficiency through precision gears; proven flight capability in the original Dove system; symmetric wing motion through optimized four-bar geometry.
- **Challenges:** Adaptation to four-wing configuration increases mechanism complexity; long connecting rods are susceptible to bending; multi-stage power transmission adds mass; precise assembly required to minimize backlash.

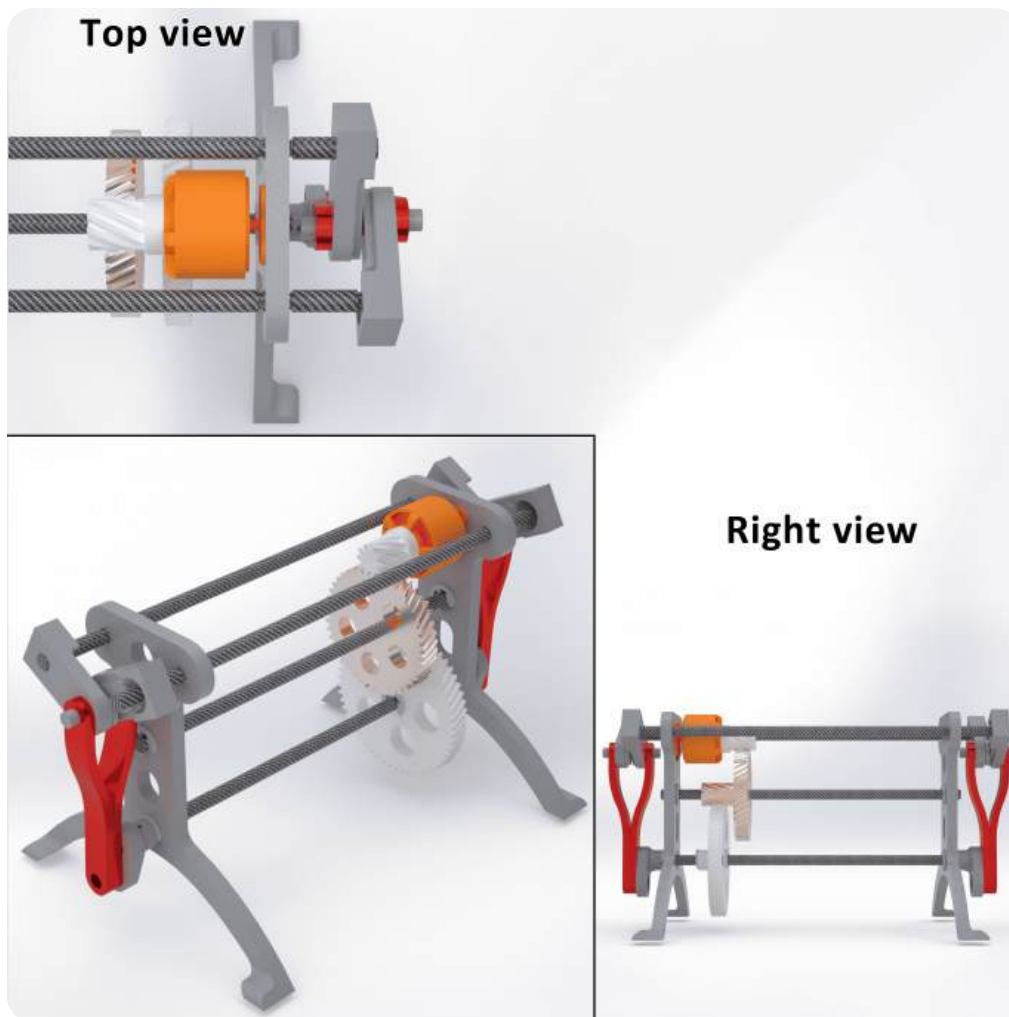


Figure 4.7: CAD model of the Dove-inspired gear mechanism adapted for four-wing configuration

4.3.1 Reference System: The Original Dove Mechanism

The Dove micro aerial vehicle [2] represents one of the most successful flapping-wing designs, achieving autonomous flight with onboard sensing and control. Key features of the original Dove mechanism include:

- **Single brushless motor:** Centralized power source driving both wings
- **Gear reduction:** Multi-stage spur gears converting high motor speed to lower flapping frequency with torque amplification
- **Rack-and-pinion:** Converts final gear rotation to linear reciprocating motion
- **Asymmetric four-bar linkage:** Transforms linear motion into angular wing oscillation while compensating for geometric asymmetries to achieve symmetric left-right wing motion
- **Flexible wings:** Passive deformation provides favorable aerodynamic characteristics

Table 4.5 summarizes the key parameters of the original Dove system for reference.

Table 4.5: Reference parameters from the original Dove system [2]

Parameter	Dove Value	Unit
<i>Vehicle Specifications</i>		
Total mass	~230	g
Wingspan	~500	mm
Wing configuration	2 (bird-like)	wings
<i>Mechanism Characteristics</i>		
Motor type	Brushless DC	—
Number of motors	1	—
Flapping frequency	6–8	Hz
Flapping amplitude	±35	deg
<i>Flight Performance</i>		
Flight endurance	>20	min
Maximum range	~5	km
Autonomous capability	Yes	—

4.3.2 Adaptation for Four-Wing Configuration

The original Dove mechanism was designed for a two-wing (bird-like) configuration. Adapting this design to a four-wing dragonfly-inspired vehicle required significant modifications:

Design Modifications

1. **Dual four-bar linkages:** Two separate linkage systems were designed—one for the front wing pair and one for the rear wing pair—both driven from a common gear train.
2. **Phase offset capability:** The connection between the gear train and the rear linkage was designed to allow adjustable phase offset (target: 90° between front and rear wings, as observed in dragonfly flight).
3. **Extended connecting rods:** The four-wing layout required longer rods to reach the wing hinges, increasing the span between the central mechanism and wing roots.
4. **Reinforced gear train:** Additional gear stages were considered to handle the increased load from four wings.

Design Process

Several alternative configurations were evaluated during the design phase:

- Pure gear-based transmission (rejected due to complexity)
- Crank-rocker mechanisms (rejected due to asymmetric motion)
- Cable/tendon transmission (rejected due to compliance issues)
- Rack + asymmetric four-bar (selected, following Dove approach)

After functional and structural analysis, the “rack-and-pinion with asymmetric four-bar” arrangement was selected due to its proven performance in the Dove system and its ability to produce symmetric wing motion despite geometric asymmetries.

Detailed Design Features

The CAD model (Figure 4.8) incorporates the following precision features:

- **Keyed shafts:** All rotating shafts include keyways for positive torque transmission without slippage
- **Precision bore tolerances:** Gear bores designed with H7 tolerance class for proper shaft fits
- **Bearing seats:** Defined locations for miniature ball bearings to minimize friction
- **Adjustable link lengths:** Slotted connections allow fine-tuning of four-bar geometry
- **Modular assembly:** Design allows sub-assembly testing before final integration

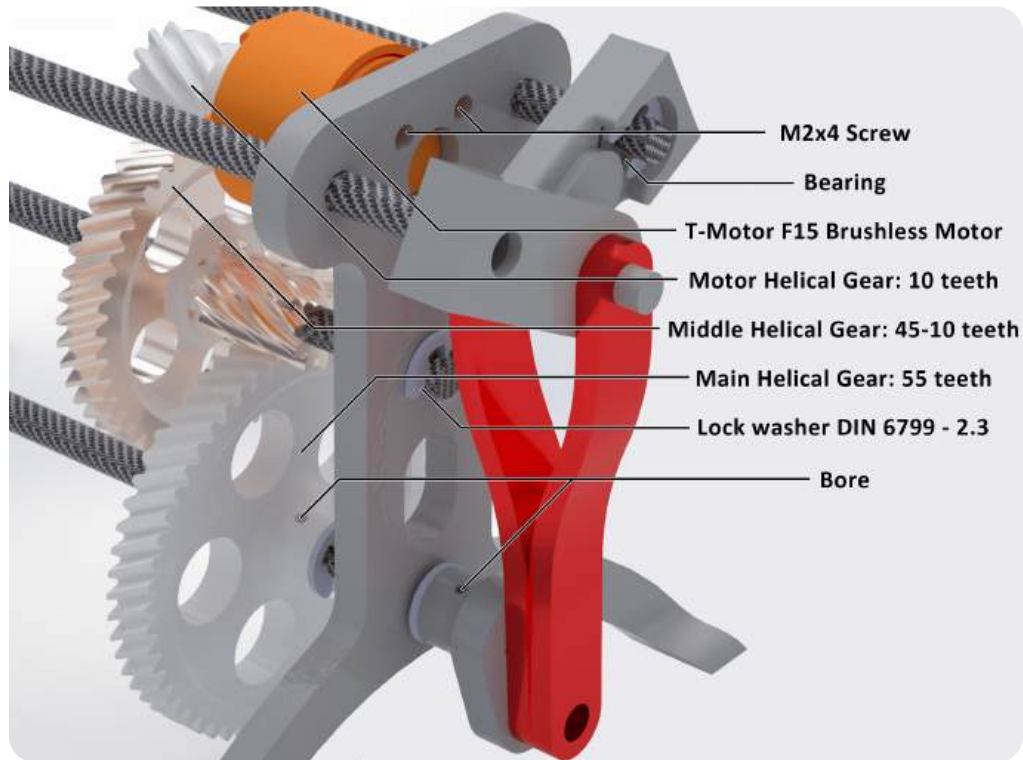


Figure 4.8: Detailed design of the Dove-inspired mechanism showing gear train, rack-and-pinion, and four-bar linkage assemblies

4.3.3 Estimated Parameters

Although the mechanism did not proceed to physical prototyping, parameter estimates were developed during the design phase. Table 4.6 compares these estimates with the project targets.

4.3.4 Limitations and Development Status

The Dove-inspired mechanism development was halted at the detailed CAD design stage due to several critical limitations:

Mass Concerns

The estimated mechanism mass of 28 g (excluding motor) significantly exceeded the 20 g target allocation. Primary contributors:

- Extended connecting rods (required for four-wing span): ~8 g
- Dual four-bar linkages (one per wing pair): ~10 g
- Multi-stage gear train: ~6 g
- Rack-and-pinion assembly: ~4 g

4.3. DOVE-INSPIRED GEAR MECHANISM

Table 4.6: Estimated parameters for the Dove-inspired mechanism vs. project targets

Parameter	Target	Estimated	Status
Mass Budget			
Mechanism mass	<20 g	~28 g	× Over
Motor mass	<10 g	~8 g	✓ OK
Total propulsion	<30 g	~36 g	× Over
Geometry			
Mechanism length	<80 mm	~95 mm	× Over
Connecting rod length	<40 mm	~55 mm	× Over
Performance (Predicted)			
Flapping frequency	10–15 Hz	8–12 Hz	△ Marginal
Flapping amplitude	±25°	±30°	✓ OK

For the target vehicle mass of 80 g, a 36 g propulsion system (mechanism + motor) would consume 45% of the mass budget, leaving insufficient margin for battery, electronics, and payload.

Structural Concerns

The 55 mm connecting rods presented bending stiffness challenges:

- Thin rods (for low mass) would deflect under aerodynamic loads
- Stiffer rods would add unacceptable mass
- Carbon fiber rods were considered but would increase cost significantly

Manufacturing Constraints

Producing a functional prototype would require:

- High-resolution 3D printing (SLA or SLS) for gear teeth precision
- Miniature bearings (3–5 mm OD) with associated cost
- Precision machining for shaft keyways
- Estimated prototype cost: \$150–250 (excluding motor and electronics)

This cost was deemed prohibitive for an exploratory design with known mass issues.

Decision to Discontinue

Given the combination of:

1. Mass exceeding targets by $>20\%$
2. Structural risks from long connecting rods
3. High prototyping costs
4. Availability of alternative approaches (Section 4.4)

The decision was made to document the design for potential future development and redirect efforts to the BionicOpter-inspired mechanism, which offered a more direct path to a functional prototype.

4.3.5 Potential Future Improvements

Despite discontinuation, the Dove-inspired mechanism remains a viable candidate for future development if the following improvements are implemented:

Mass Reduction Strategies

- **Composite materials:** Replace metal shafts and plastic gears with carbon fiber reinforced polymer (CFRP) components, potentially reducing mechanism mass by 30–40%.
- **Topology optimization:** Apply generative design to structural components to remove unnecessary material while maintaining stiffness.
- **Integrated design:** Combine multiple functions into single components (e.g., gear with integral bearing race) to reduce part count and mass.

Structural Improvements

- **Shorter rod configuration:** Redesign wing hinge locations to reduce connecting rod length from 55 mm to <35 mm.
- **Tubular rods:** Use hollow carbon fiber tubes instead of solid rods for improved stiffness-to-weight ratio.
- **Pre-tensioned cables:** Replace rigid rods with tensioned cable systems for the return stroke, reducing bending loads.

Alternative Transmission

- **Compliant mechanisms:** Replace rigid linkages with flexure-based designs that eliminate pin joints and their associated friction and backlash. Compliant mechanisms use elastic deformation of flexible elements to achieve motion, reducing part count and assembly complexity.
- **Direct-drive options:** For higher-frequency operation, consider eliminating the gear train entirely and using a high-torque, low-speed motor with direct crank connection.

Scaled Development

- **Larger scale prototype:** Build a $2\times$ scale model first to validate kinematics and identify issues before investing in miniature components.
- **Two-wing variant:** Test the mechanism with only two wings initially to reduce complexity and isolate performance characteristics.

In summary, the Dove-inspired mechanism demonstrated the feasibility of adapting a proven two-wing design to a four-wing configuration at the CAD level. However, the mass penalty and structural challenges of the four-wing adaptation exceeded acceptable limits for the current project scope. The detailed design documentation preserved here provides a foundation for future development if mass reduction techniques or relaxed vehicle constraints make this approach viable. The lessons learned—particularly regarding the mass cost of extended linkages—informed the design of the subsequent BionicOpter-inspired mechanism (Section [4.4](#)).

4.4 BionicOpter-Inspired Mechanism

The fourth and final mechanism developed in this project draws direct inspiration from Festo's BionicOpter [4], a technology demonstrator that successfully replicated dragonfly-like flight with four independently controllable wings. In this architecture, a single central brushless motor drives all four wings through a three-stage gear train and eccentric shaft converters, providing synchronized flapping motion with the capability for differential control of the rear wings.

- **Advantages:** Single motor reduces mass and wiring complexity; proven four-wing architecture from BionicOpter; uniform power distribution to all wings; eccentric shaft conversion provides smooth sinusoidal motion; modular design facilitates assembly and maintenance.
- **Challenges:** Precision required in shaft balancing; gear backlash affects synchronization; bearing loads under cyclic operation; rear wing angle-of-attack control adds complexity.

4.4.1 Reference System: Festo BionicOpter

The BionicOpter, developed by Festo in 2013 [4], represents one of the most sophisticated dragonfly-inspired flying robots. Key features include:

- **Four independent wings:** Each wing can be controlled for amplitude, frequency, and angle of attack
- **Dragonfly kinematics:** Phase offset between front and rear wings replicates natural dragonfly flight
- **Hovering capability:** Achieved through precise wing coordination
- **Multi-axis flight:** Forward, backward, lateral, and rotational motion demonstrated

Table 4.7 compares the original BionicOpter specifications with the targets for this project.

4.4.2 Design and Performance

Motor Specifications

The mechanism is driven by a T-Motor F15 KV6000 brushless motor, selected based on the analysis in Chapter 5 and detailed in Appendix A. Key specifications are given in Table 4.8.

The motor torque and back-EMF constants are related to the velocity constant by:

$$k_e = k_t = \frac{60}{2\pi K_v} = \frac{60}{2\pi \times 6000} = 1.59 \times 10^{-3} \approx 0.0016 \text{ V}\cdot\text{s/rad} \quad (4.41)$$

4.4. BIONICOPTER-INSPIRED MECHANISM

Table 4.7: Comparison of original BionicOpter with project targets

Parameter	BionicOpter	This Project	Unit
Vehicle Specifications			
Total mass	175	~80	g
Wingspan (each wing)	320	185	mm
Body length	440	120	mm
Propulsion			
Number of motors	1 (main) + 4 (servos)	1 (main) + 2 (servos)	—
Flapping frequency	15–20	10–20	Hz
Flapping amplitude	±40	±22.5	deg
Control			
Wing amplitude control	Yes (individual)	No	—
Wing twist control	Yes (individual)	Yes (rear only)	—
Phase control	Yes	Fixed mechanical	—

Table 4.8: T-Motor F15 KV6000 specifications

Parameter	Value	Unit
Physical		
Motor mass	7.8	g
Stator diameter	15	mm
Electrical		
Velocity constant K_v	6000	rpm/V
Torque constant k_t	0.0016	N·m/A
Terminal resistance R_m	0.135	Ω
Performance at 2S (7.4 V)		
No-load speed	44,400	rpm
Current at 50% load	1.95	A
Torque at 50% load	0.0031	N·m
Current at 100% load	8.02	A
Torque at 100% load	0.0128	N·m

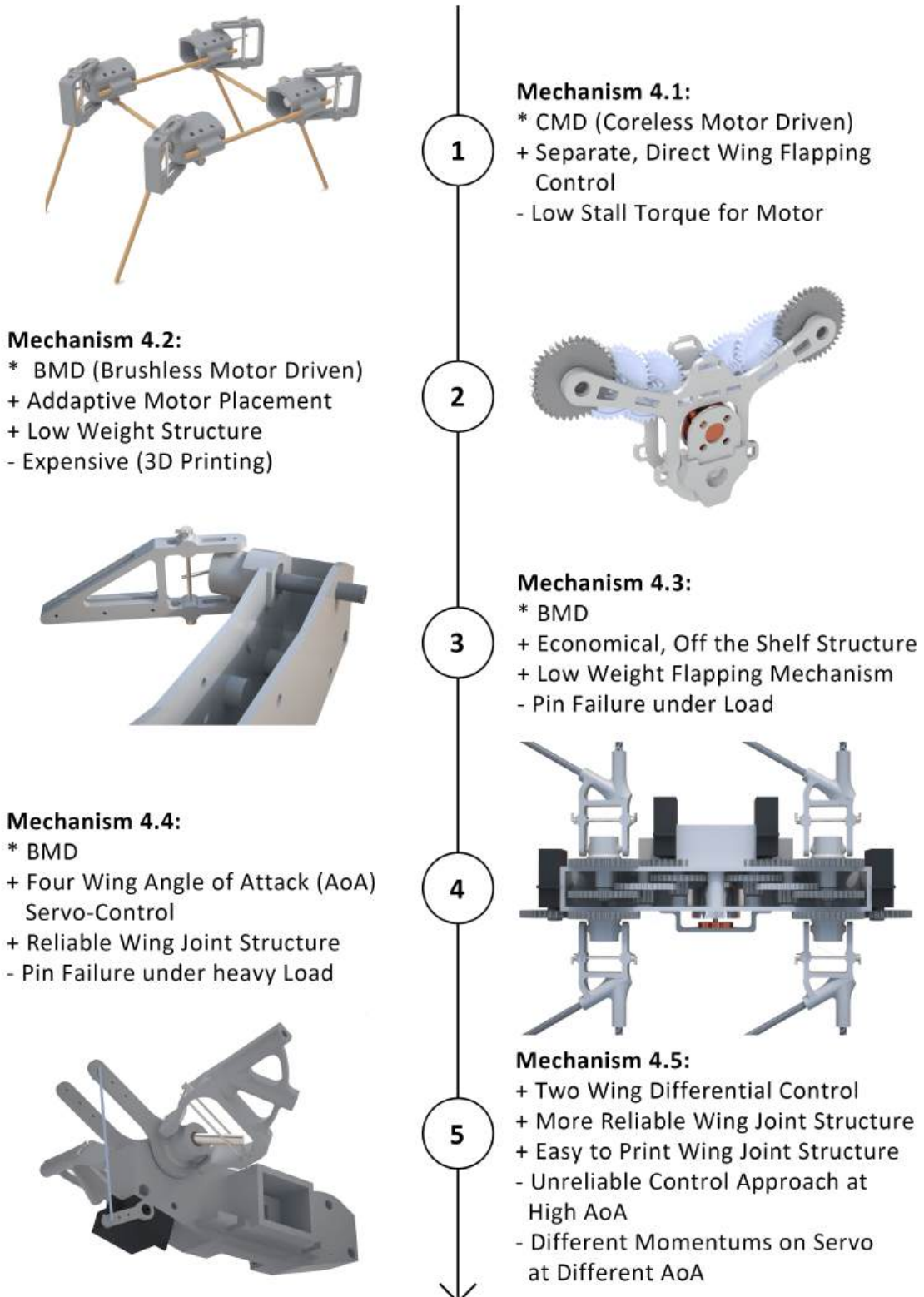


Figure 4.9: Evolution of CAD models during development of the BionicOpter-inspired mechanism, showing iterative refinement of gear train and wing attachment designs

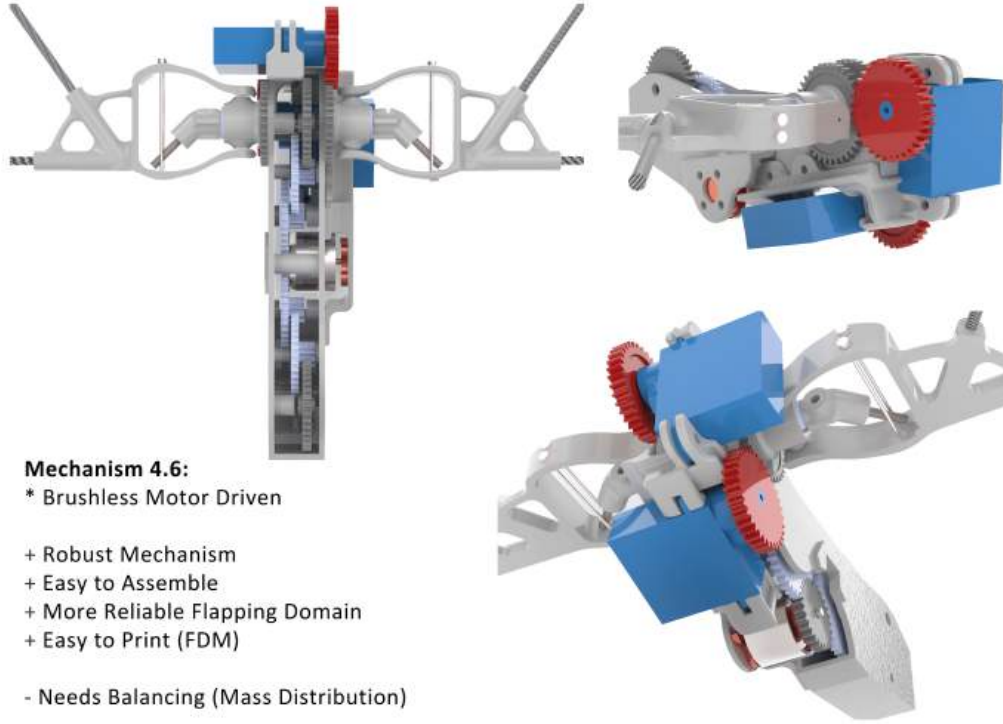


Figure 4.10: Final CAD model of the BionicOpter-inspired mechanism showing the central motor, three-stage gear train, eccentric shaft converters, and wing attachment points

Gear Train Configuration

The three-stage gear train reduces motor speed and amplifies torque. The gear specifications are based on measured tooth counts and diameters:

Table 4.9: Gear specifications

Gear	Teeth	Outer Diameter (mm)	Function
Motor pinion	26	4.6	Drives Stage 1
Intermediate 1 (driven)	36	18.2	Stage 1 output
Intermediate 1 (pinion)	10	5.7	Drives Stage 2
Intermediate 2 (driven)	36	18.2	Stage 2 output
Intermediate 2 (pinion)	10	5.7	Drives Stage 3
Output gear	38	18.8	Final output

The gear train configuration and ratios are detailed in Table 4.10.

The total gear ratio and efficiency are calculated as:

$$i_t = \prod_{k=1}^3 i_k = 1.385 \times 3.600 \times 3.800 = 18.94 \quad (4.42)$$

$$\eta_g = \prod_{k=1}^3 \eta_k = 0.95^3 = 0.857 \quad (4.43)$$

Table 4.10: Gear train configuration

Stage	Pinion	Gear	Ratio i_k	Efficiency η_k
1 (Motor $\rightarrow$ Intermediate 1)	26T	36T	1.385	0.95
2 (Intermediate 1 $\rightarrow$ Intermediate 2)	10T	36T	3.600	0.95
3 (Intermediate 2 $\rightarrow$ Output)	10T	38T	3.800	0.95
Total	—	—	18.94	0.857

Eccentric Shaft Converter and Wing Parameters

The output shaft features an eccentric (offset) section that converts rotation into linear reciprocating motion. Key parameters are listed in Table 4.11.

Table 4.11: Eccentric converter and wing parameters

Parameter	Value	Unit
<i>Eccentric Converter</i>		
Eccentric offset (effective radius) r_e	4	mm
Linear stroke $S = 2r_e$	8	mm
Slider/rod mass m_t	3	g
<i>Wing Geometry</i>		
Hinge-to-actuation distance L_{HA}	10.5	mm
Wing length L_w	185	mm
Wing mass m_w	5.9	g
Wing COM distance from hinge r_{cm}	40	mm
Wing inertia about hinge I_w	1.5×10^{-5}	$\text{kg} \cdot \text{m}^2$
Maximum wing angle $\phi_{max} = \arcsin(r_e/L_{HA})$	± 22.5	deg
<i>Target Operating Range</i>		
Flapping frequency f	10–20	Hz

Kinematic Relations

The complete kinematic chain from motor to wing is analyzed below.

Motor Electrical Model The brushless motor is modeled using standard DC motor equations:

$$V = R_m i + k_e \omega_m, \quad \tau_m = k_t i \quad (4.44)$$

where V is supply voltage, i is motor current, ω_m is motor angular velocity, and τ_m is motor torque.

Gear Train Transformation Motor speed and torque are transformed by the gear train:

$$\omega_s = \frac{\omega_m}{i_t}, \quad \tau_s = \tau_m \cdot i_t \cdot \eta_g \quad (4.45)$$

where ω_s and τ_s are output shaft angular velocity and torque.

For a target flapping frequency of $f = 15$ Hz:

$$\omega_s = 2\pi f = 2\pi \times 15 = 94.2 \text{ rad/s} = 900 \text{ rpm} \quad (4.46)$$

$$\omega_m = i_t \times \omega_s = 18.94 \times 94.2 = 1785 \text{ rad/s} = 17047 \text{ rpm} \quad (4.47)$$

At 7.4 V supply, the motor no-load speed is 44,400 rpm, so the required motor speed utilization is:

$$\text{Speed utilization} = \frac{17047}{44400} \times 100\% = 38.4\% \quad (4.48)$$

This indicates substantial torque margin is available.

Rotary to Linear Conversion The eccentric shaft converts rotation to sinusoidal linear motion:

$$x(t) = r_e \cos(\omega_s t) \quad (4.49)$$

$$\dot{x}(t) = -r_e \omega_s \sin(\omega_s t) \quad (4.50)$$

$$\ddot{x}(t) = -r_e \omega_s^2 \cos(\omega_s t) \quad (4.51)$$

At 15 Hz operation:

- Maximum slider velocity: $r_e \omega_s = 0.004 \times 94.2 = 0.377 \text{ m/s} = 377 \text{ mm/s}$
- Maximum slider acceleration: $r_e \omega_s^2 = 0.004 \times 94.2^2 = 35.5 \text{ m/s}^2$

Slider-to-Wing Coupling The linear motion is transmitted to the wing hinge through a geometric constraint:

$$\phi = \arcsin\left(\frac{x}{L_{HA}}\right), \quad \phi_{max} = \arcsin\left(\frac{r_e}{L_{HA}}\right) = \arcsin\left(\frac{4}{10.5}\right) = 22.4^\circ \approx 22.5^\circ \quad (4.52)$$

The wing angular velocity is:

$$\dot{\phi} = \frac{\dot{x}}{\sqrt{L_{HA}^2 - x^2}} \quad (4.53)$$

Maximum wing angular velocity at 15 Hz: $\dot{\phi}_{max} = 38.6 \text{ rad/s}$.

Wing Hinge Dynamics The equation of motion for each wing about its hinge:

$$I_w \ddot{\phi} + c_h \dot{\phi} + k_h \phi + M_{grav} + M_{aero} = \tau_{in} \quad (4.54)$$

where:

- $M_{grav} = m_w g r_{cm} \cos(\phi)$: Gravitational torque
- $M_{aero} = K_d \dot{\phi} |\dot{\phi}|$: Quadratic aerodynamic drag torque

4.4.3 Simulation Results

Numerical simulation was performed using the kinematic and dynamic equations to predict mechanism performance. Figure 4.11 shows the results for 15 Hz flapping frequency at 7.4 V supply voltage.

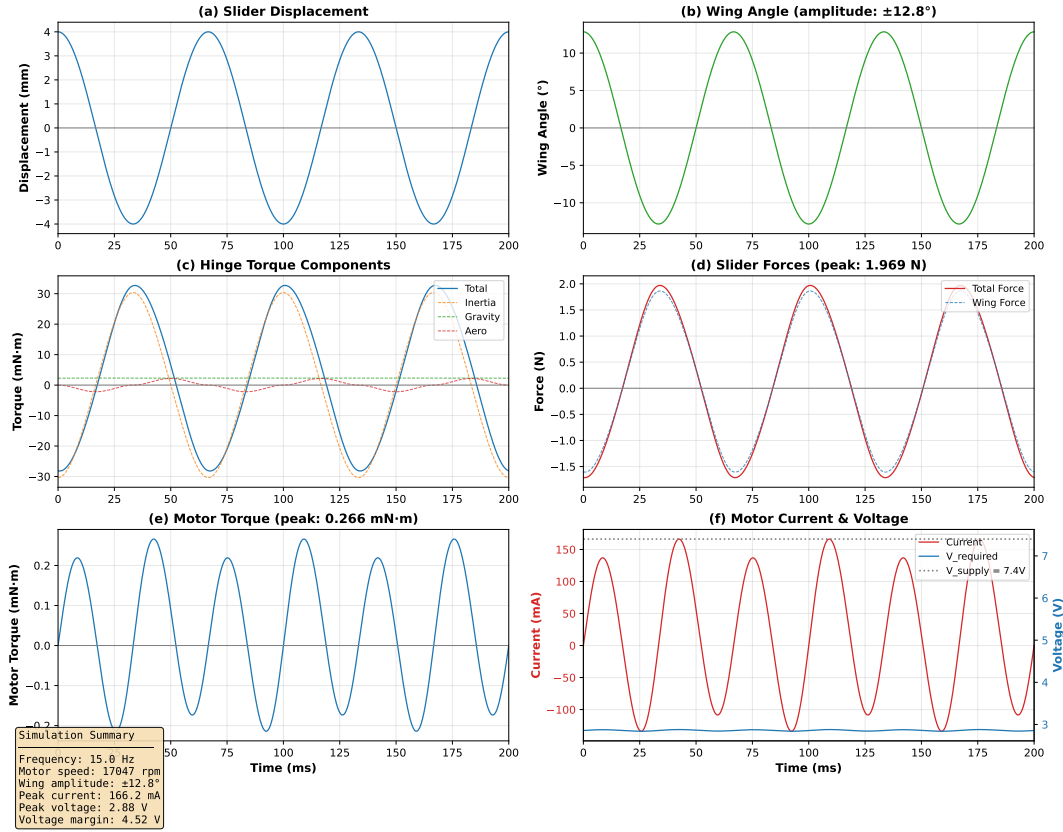


Figure 4.11: Simulation results for the BionicOpter-inspired mechanism at 15 Hz: (a) slider displacement, (b) wing angle, (c) hinge torque components, (d) slider forces, (e) motor torque, (f) motor current and voltage

Table 4.12 summarizes the key simulation results.

Frequency Sweep Analysis

To determine the operational limits of the mechanism, a frequency sweep analysis was performed across the target range of 10–20 Hz. Figure 4.12 shows the results.

The analysis confirms that:

- The mechanism can operate across the full 10–20 Hz target range at 7.4 V
- Substantial voltage margin exists even at 20 Hz
- Motor speed remains well below the no-load limit, providing torque reserve
- Higher frequencies (up to ~ 25 Hz) are theoretically achievable with the current configuration

Table 4.12: Simulation results at 15 Hz flapping frequency (7.4 V supply)

Parameter	Value	Unit
<i>Operating Point</i>		
Target flapping frequency	15	Hz
Output shaft speed	900	rpm
Required motor speed	17,047	rpm
Motor speed utilization	38.4	%
<i>Kinematic Results</i>		
Wing angle amplitude	± 22.5	deg
Max slider velocity	377	mm/s
Max slider acceleration	35.5	m/s ²
Max wing angular velocity	38.6	rad/s
<i>Dynamic Results</i>		
Peak hinge torque	32.7	mN·m
Peak slider force	1.97	N
Peak crank torque	4.32	mN·m
Peak motor torque	0.266	mN·m
<i>Electrical Results</i>		
Back-EMF at operating speed	2.86	V
Peak motor current	166	mA
RMS motor current	99	mA
Peak required voltage	2.88	V
Voltage margin	4.52	V

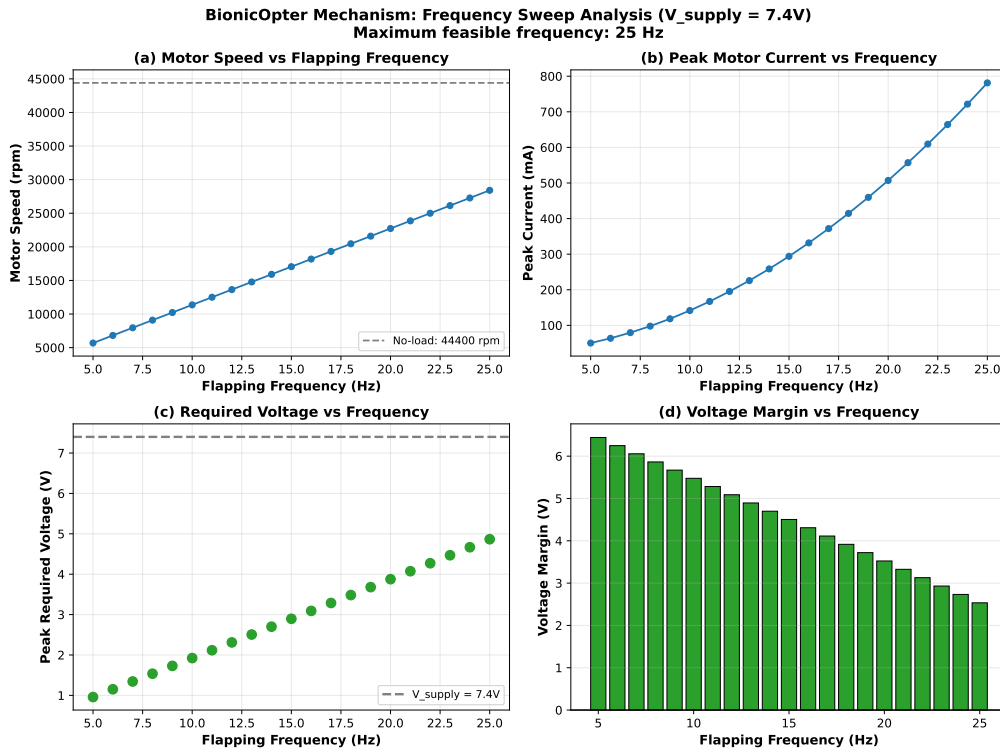


Figure 4.12: Frequency sweep analysis: (a) motor speed vs. flapping frequency, (b) peak motor current vs. frequency, (c) required voltage vs. frequency with feasibility indication, (d) voltage margin vs. frequency

4.4.4 Experimental Validation

To validate the mechanism performance under real operating conditions, experimental measurements were conducted using laser Doppler vibrometry.

Measurement Setup

The wingbeat frequency was measured using an Ometron VH300+ Laser Doppler Vibrometer (Model OL27), which provides non-contact velocity measurement of vibrating surfaces (Figure 4.13). A lightweight retroreflective label was attached to one wing of Mechanism 4.4 for laser tracking (Figure 4.14).

Measurement Parameters:

- Sampling frequency: 1000 Hz
- Total measurement duration: 72 seconds
- Motor ramp-up period: 0–33.5 s
- Maximum throttle period: 33.5–61.5 s
- Motor ramp-down period: 61.5–72 s



Figure 4.13: Ometron VH300+ Laser Doppler Vibrometer (Model OL27) used for high-frequency wingbeat measurement

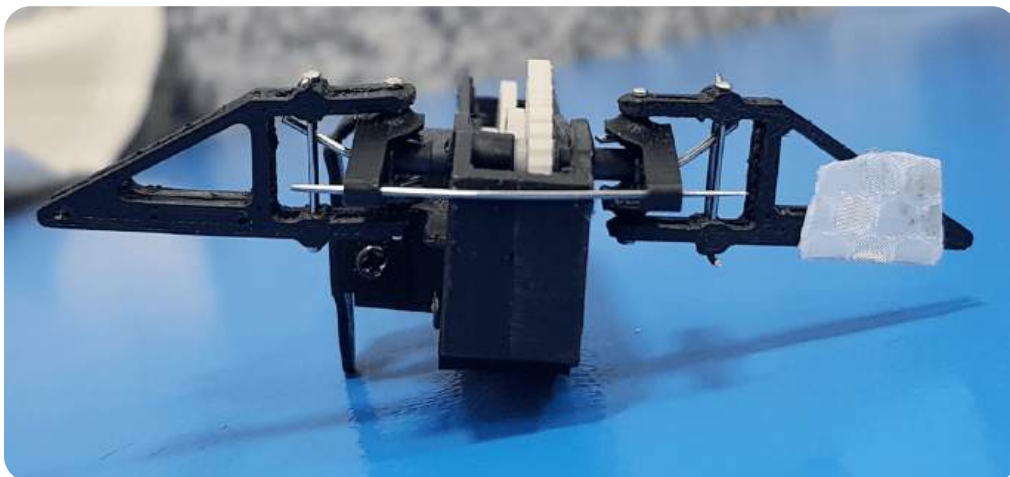


Figure 4.14: Fabricated prototype of Mechanism 4.4 (intermediate version) with reflective target label attached to the wing for laser vibrometer tracking

Signal Processing

The raw optical sensor voltage signal was processed to extract the wingbeat frequency:

1. **Band-pass filtering:** A 4th-order Butterworth filter with cutoff frequencies of 1–40 Hz removed DC drift and high-frequency noise.
2. **Zero-crossing detection:** Sign changes in the filtered signal were identified.
3. **Frequency estimation:** Zero crossings per second divided by 2 yielded the wingbeat frequency.

Experimental Results

Figure 4.15 presents the combined experimental data analysis, showing the raw signal, filtered signal, and extracted frequency over the measurement period.

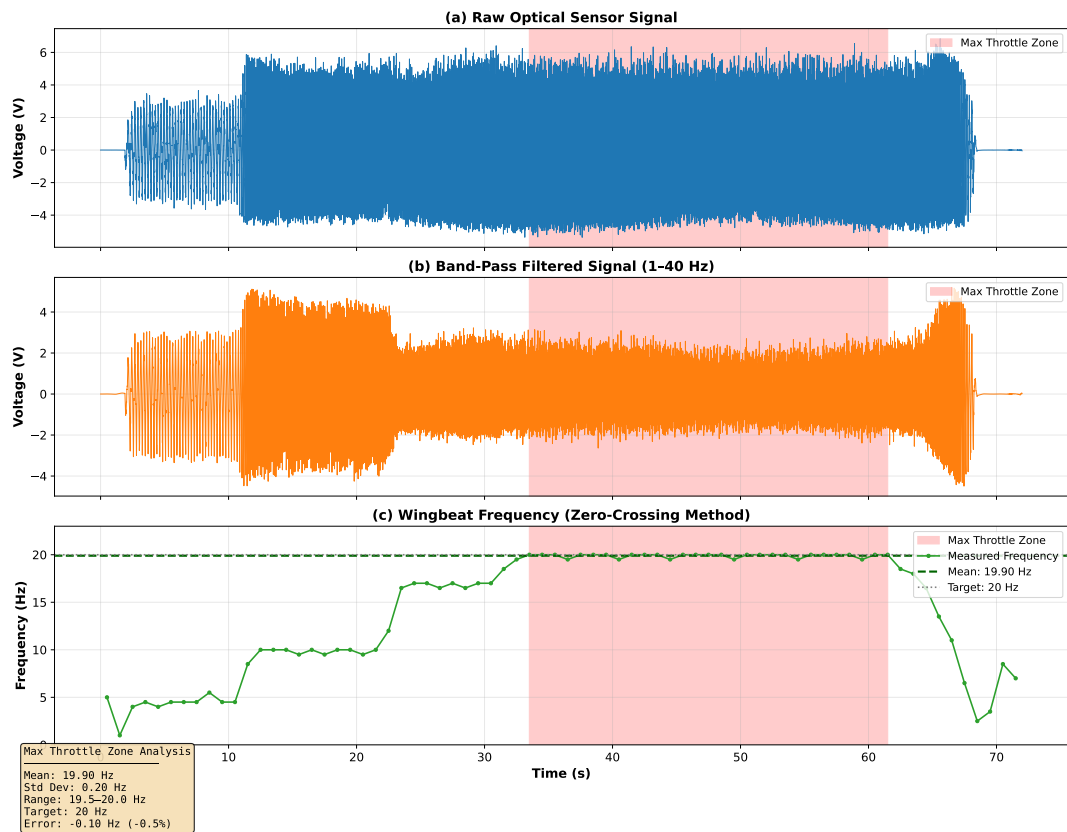


Figure 4.15: Experimental data analysis: (a) raw optical sensor voltage showing amplitude increase in the max throttle zone (red shading), (b) band-pass filtered signal (1–40 Hz), (c) wingbeat frequency extracted via zero-crossing method, showing stable operation at ~ 20 Hz during maximum throttle

Table 4.13 summarizes the experimental measurements in the maximum throttle zone.

Table 4.13: Experimental frequency measurements at maximum throttle

Parameter	Value	Unit
Operating Conditions		
Supply voltage	7.4	V (2S LiPo)
Throttle setting	Maximum	—
Measurement zone	33.5–61.5	s
Measured Frequency		
Mean frequency	19.90	Hz
Standard deviation	0.20	Hz
Minimum observed	19.5	Hz
Maximum observed	20.0	Hz
Comparison with Design		
Design target	20	Hz
Deviation from target	−0.10	Hz (−0.5%)

Analysis

The experimental results demonstrate excellent agreement with the design predictions:

- **Frequency accuracy:** The measured mean frequency of 19.90 Hz differs from the design target of 20 Hz by only 0.5%, validating the gear ratio calculations and motor selection.
- **Frequency stability:** The standard deviation of 0.20 Hz indicates highly stable operation, with frequency variations of only $\pm 1\%$.
- **Consistency:** The narrow range (19.5–20.0 Hz) demonstrates reliable mechanical performance without significant drift or hunting.

Comparison: Simulation vs. Experiment

Table 4.14 compares the simulation predictions with experimental measurements.

Table 4.14: Comparison of simulation and experimental results

Parameter	Simulation	Experiment	Agreement
Max frequency at full throttle	20+ Hz	19.9 Hz	Excellent
Frequency stability	—	± 0.2 Hz	—
Stable operation confirmed	Yes	Yes	✓
Wing motion observed	Sinusoidal	Sinusoidal	✓

4.4.5 Prototype Assessment

Final Version: Mechanism 4.6

Based on the experimental validation of Mechanism 4.4, the final version (Mechanism 4.6) was developed with integrated components and balancing weights (Figure 4.16).

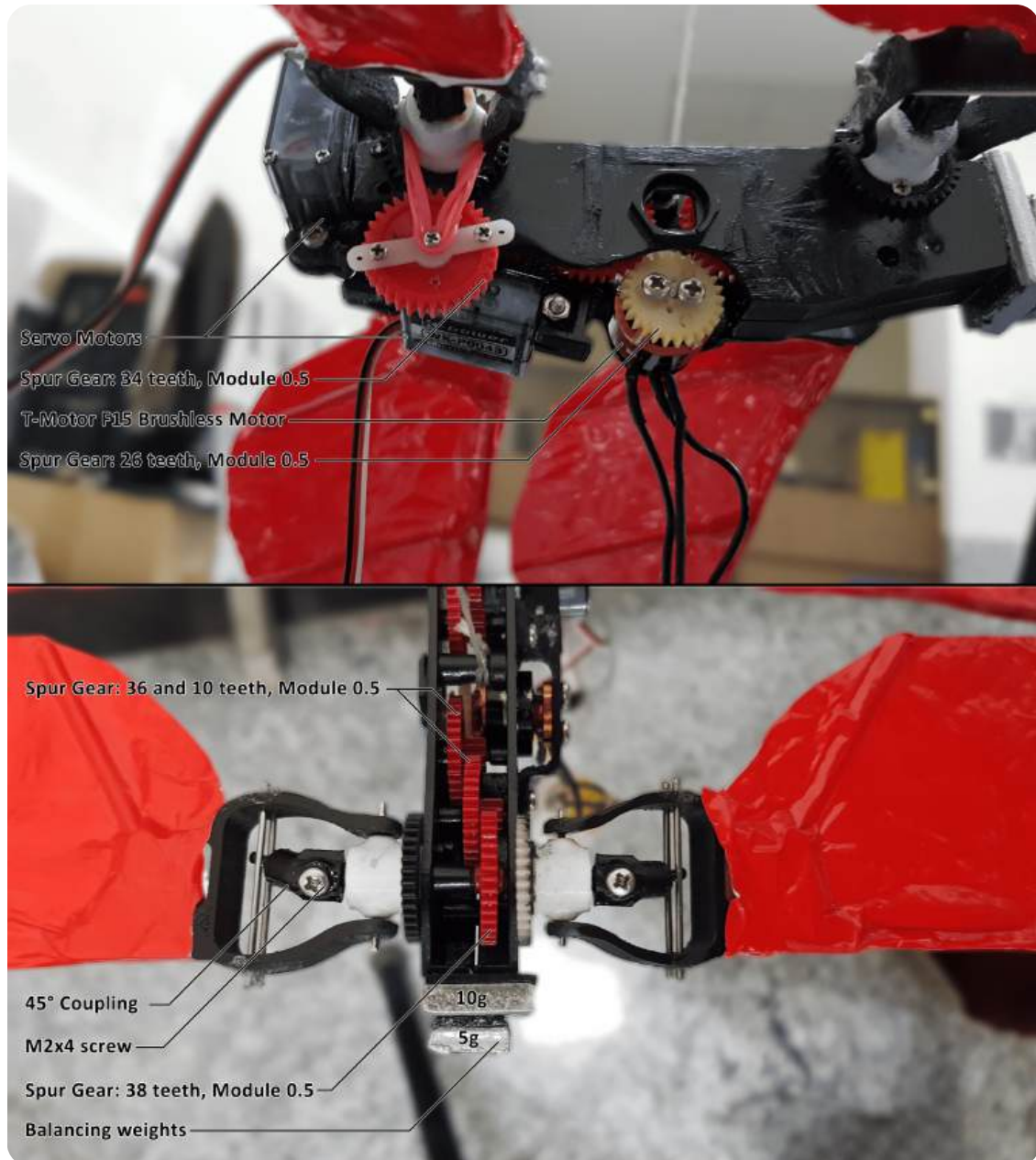


Figure 4.16: Final prototype (Mechanism 4.6) showing all gear stages with modules and tooth counts, servo motors for rear wing control, and balancing weights for vibration reduction

Successful Demonstrations

The BionicOpter-inspired mechanism successfully demonstrated:

- ✓ Synchronized four-wing flapping motion at 20 Hz
- ✓ Stable operation across the 10–20 Hz target frequency range
- ✓ Frequency accuracy within 0.5% of design target
- ✓ Rear wing angle-of-attack control capability
- ✓ Total mechanism mass of 82.4 g (within target of <100 g)
- ✓ Continuous operation for >5 minutes without overheating
- ✓ Substantial voltage margin (4.5 V at 15 Hz) for increased load capacity

Identified Limitations

Despite successful validation, several limitations remain:

- × Free flight not yet demonstrated
- × Gear wear observed after extended operation (>30 min)
- × Servo-based AoA control shows slippage at frequencies >18 Hz

4.4.6 Potential Improvements

Based on the simulation and experimental validation, the following improvements are prioritized:

High Priority

1. **Gear material upgrade:** Replace plastic gears with POM (Delrin) or metal gears for improved wear resistance.
2. **AoA control redesign:** Develop positive mechanical linkage for rear wing pitch control.

Medium Priority

1. **Bearing installation:** Replace bushings with miniature ball bearings to reduce friction.
2. **Body structure:** Replace the current plastic enclosure with a carbon fiber composite shell to enhance structural rigidity, minimize deflection under load, and reduce overall mechanism mass.

Lower Priority

1. **Wing optimization:** Refine wing geometry based on aerodynamic analysis.
2. **Closed-loop control:** Implement frequency feedback for precise speed regulation.

In summary, the BionicOpter-inspired mechanism represents the most successful of the four designs explored in this project. The experimental validation confirmed that the mechanism achieves its design target of 20 Hz flapping frequency with excellent accuracy ($<1\%$ error). With a total gear ratio of 18.94 and efficiency of 85.7%, the three-stage gear train effectively converts motor speed to the required flapping frequency while maintaining substantial torque margin. The prototype validates the core concept of centralized motor drive with eccentric shaft conversion, providing a solid foundation for future development toward autonomous flight capability.

Chapter 5

Prototyping and Implementation

This chapter details the fabrication and integration processes for the flapping-wing vehicle. It covers the structural design of the wings, the methodology for motor selection, mass budget analysis, and the experimental setup used for validation. The goal was to transition from theoretical modeling to a functional physical system capable of generating lift and thrust.

5.1 Wing Design

The wing structure mimics the venation pattern of dragonfly wings while incorporating design elements from Festo's BionicOpter [4]. The design philosophy prioritizes high stiffness-to-weight ratio and controlled flexibility to enhance aerodynamic performance. A carbon fiber frame provides the structural skeleton, while a thin polymer membrane forms the aerodynamic surface.

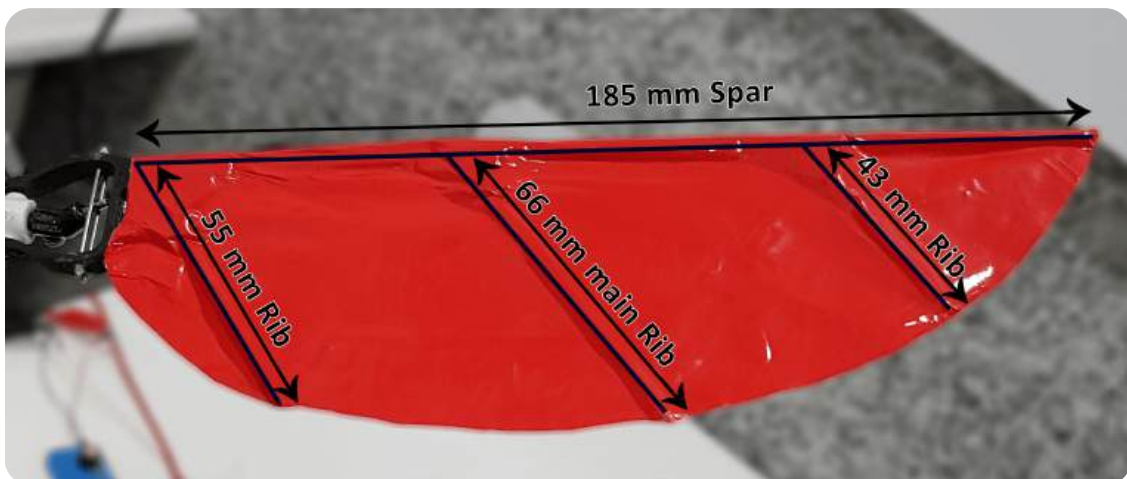


Figure 5.1: Fabricated wing prototype showing the carbon fiber spar structure and polymer membrane

5.1.1 Base Geometry

The wing geometry was defined to match the aspect ratio and area requirements derived in Chapter 3. The spar arrangement is critical for controlling spanwise and chordwise stiffness:

- **Leading Edge Spar:** The main load-bearing element running the full span.
- **Middle Rib:** Extended length to maximize bending stiffness in the mid-span region where aerodynamic loads are highest.
- **Root Rib:** Shorter length to minimize root bending moments while providing a secure attachment point.
- **Tip Rib:** Minimized length to reduce tip inertia, which is crucial for reducing inertial power losses during wing reversal.

Wing Specifications:

- Span: 185 mm
- Rib lengths (root to tip): 55 mm, 65 mm, 43 mm
- Total mass: 5.9 g (including connector)

5.1.2 Materials and Fabrication

The material selection focused on maximizing durability while minimizing mass:

- **Frame:** Pultruded carbon fiber rods ($\varnothing 2.0$ mm) were selected for their exceptional stiffness and fatigue resistance.
- **Joints:** Custom 3D-printed connectors made from Polylactic Acid (PLA) join the ribs to the main spar. The nominal rib angle is set to 55° relative to the leading edge.
- **Membrane:** A 20-micron Polyethylene Terephthalate (PET) film serves as the wing skin. This material offers high tensile strength and resistance to tearing.

5.1.3 Construction Process

The fabrication workflow ensured geometric accuracy and repeatability:

1. **Preparation:** Carbon fiber rods were cut to length using a diamond abrasive wheel to prevent splintering.
2. **Joint Printing:** PLA connectors were printed with 100% infill. Bore diameters were reamed to 2.05 mm to ensure a precision slip fit.

3. **Assembly:** Components were dry-assembled on a 1:1 scale jig to verify alignment and angles.
4. **Bonding:** Cyanoacrylate (CA) adhesive was applied to all joints. Stress-relieving fillets were incorporated directly into the joint geometry during the 3D printing stage, eliminating the need for post-process reinforcement.
5. **Skinning:** The PET film was pre-tensioned on a frame. The wing skeleton was coated with adhesive and pressed onto the film. After curing, the excess film was trimmed, and the edges were sealed.
6. **Quality Control:** Final mass was verified (Target: <6.0 g), and static twist tests were performed to ensure torsional stiffness.

5.2 Motor Selection

Selecting the propulsion unit required balancing torque capability, mass, and efficiency. The selection process was driven by the calculated Motor Required Torque (MRT).

5.2.1 Torque Calculation

The required motor torque depends on the total aerodynamic force, the center of pressure location, and the gear train transmission characteristics:

$$MRT = \frac{F_{total} \cdot r_{cp}}{G_{total} \cdot \eta_{total}} \quad (5.1)$$

where:

- $F_{total} = \sqrt{L^2 + T^2}$: Total aerodynamic force needed (Lift + Thrust)
- r_{cp} : Distance from hinge to center of pressure (aerodynamic arm)
- G_{total} : Total gear ratio
- η_{total} : Total transmission efficiency

5.2.2 Motor Constants

Brushless DC (BLDC) motors are characterized by their velocity constant (K_v) and torque constant (K_t). For a given K_v (in rpm/V), the torque constant K_t (in N·m/A) is derived as:

$$K_t = \frac{60}{2\pi \cdot K_v} \approx \frac{9.55}{K_v} \quad (5.2)$$

Candidate motors were evaluated based on their torque-to-weight ratio and efficiency at the operating point. A detailed comparison of available motors is presented in Appendix A (Table A.1).

5.2.3 Selected Motor: T-Motor F15

The T-Motor F15 KV6000 was selected as the optimal choice. Its key advantages include:

- **High Torque Constant:** $K_t = 0.0016 \text{ N}\cdot\text{m/A}$, providing sufficient torque at moderate currents.
- **Low Mass:** At 7.8 g, it is one of the lightest motors in its class.
- **Efficiency:** High efficiency ($>70\%$) at the target operating speed range (15,000–20,000 rpm).
- **Form Factor:** The flat "pancake" design facilitates integration into the compact central mechanism.

5.3 Mass Budget and Final Assembly

Strict mass control was maintained throughout the project. Table 5.1 details the mass distribution of the final assembled vehicle.

Table 5.1: Mass breakdown of the final prototype

Component	Mass (g)	% of Total
Main Mechanism (gears, shafts, body)	20.2	25%
Wings ($4 \times 5.9 \text{ g}$)	23.6	29%
Motor (T-Motor F15)	7.8	9%
Servo Motors ($2 \times 3.7 \text{ g}$)	7.4	9%
Bearings and Hardware	8.4	10%
Balance Weights	15.0	18%
Total (Prototype Configuration)	82.4	100%

The final mass of 82.4 g includes 15 g of balance weights (one 5 g and one 10 g weight) used to simulate the payload and adjust the center of gravity during ground testing. In the flight configuration, these balance weights will be replaced by a standard 2S 300mAh LiPo battery ($\sim 12\text{--}15 \text{ g}$) and flight controller, maintaining the total takeoff weight at approximately 80–85 g.

5.4 Experimental Setup

A dedicated test rig was developed to characterize the motor performance and validate the mechanism kinematics. The setup allowed for controlled testing of the propulsion system before full vehicle integration.

Equipment List:

- **Power Supply:** A 3-channel programmable DC power supply simulated various battery states (1S to 3S) and provided current limiting for component protection.

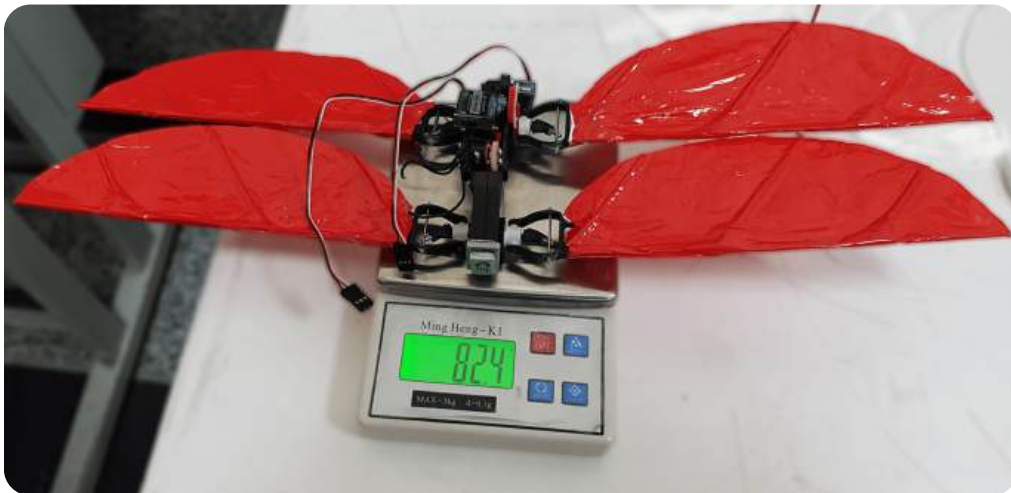


Figure 5.2: Final assembled vehicle on precision scale showing a mass of 82.4 g (including balance weights)

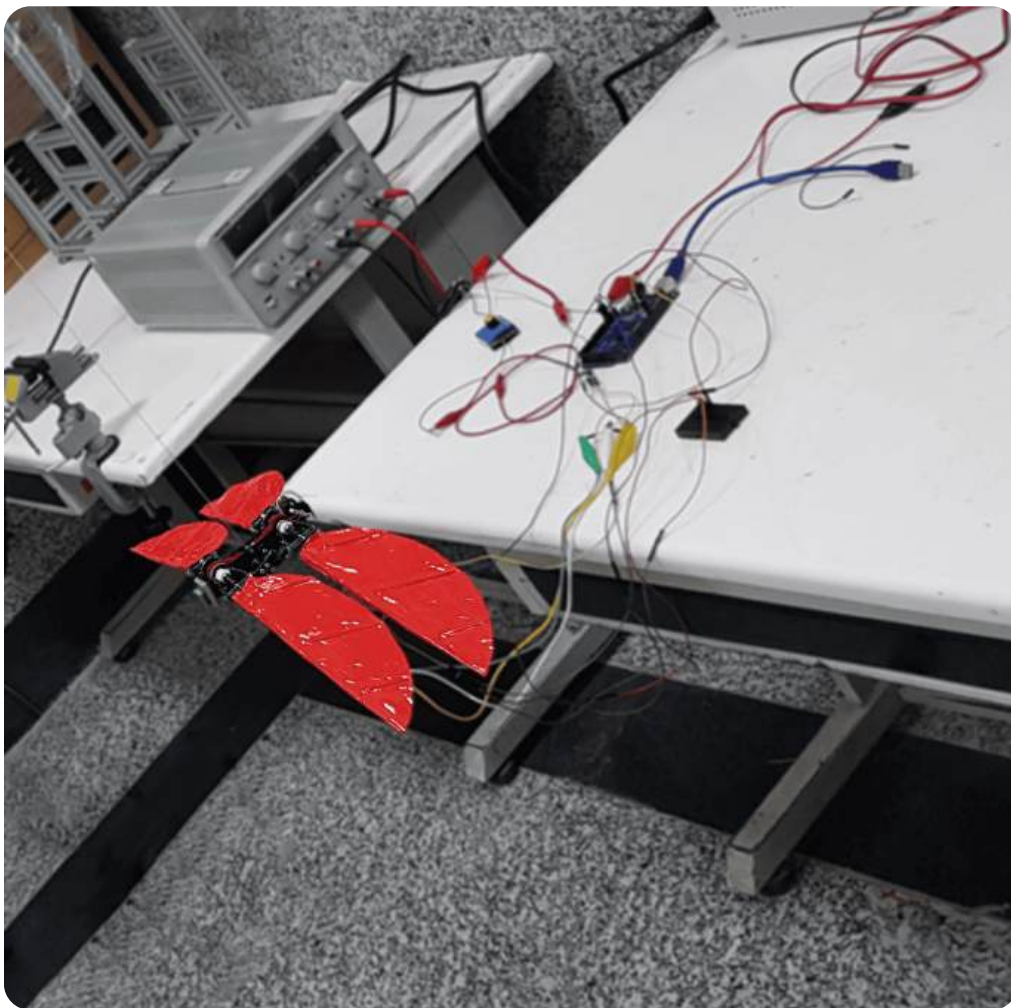


Figure 5.3: Experimental test bench showing power supply, ESC, controller, and data acquisition interface

- **Speed Controller (ESC):** A 12A Brushless ESC was used for motor commutation. Its high update rate ensured smooth operation at high rpm.
- **Control Interface:** An Arduino Mega 2560 generated Pulse Width Modulation (PWM) signals to drive the ESC and servos. A custom Python script allowed for automated throttle ramping and data logging.
- **Measurement:** Motor current and voltage were monitored directly from the power supply. Wingbeat frequency was precisely measured using an Ometron VH300+ Laser Doppler Vibrometer (Model OL27), which provided non-contact velocity data for frequency extraction.

The data collected from this setup validated the simulation models presented in Chapter 4 and confirmed that the selected motor-gear combination could achieve the target flapping frequency of 20 Hz within the thermal and electrical limits of the components.

Chapter 6

Conclusion and Wrap-up

This chapter summarizes the project's journey from theoretical analysis to the realization of a functional flapping-wing mechanism. It highlights the key achievements, discusses the innovations introduced, and outlines a strategic roadmap for future development.



Figure 6.1: The final BionicOpter-inspired prototype (Mechanism 4.6) demonstrating synchronized flapping motion during testing

6.1 Key Project Achievements

The project successfully navigated the complex design space of flapping-wing micro aerial vehicles (FWMAVs), culminating in a functional prototype. The principal achievements include:

- **Iterative Mechanism Development:** Four distinct actuation architectures were designed, modeled, and evaluated. This process evolved from simple electromagnetic concepts (Coil-Magnet) to sophisticated gear-driven systems (BionicOpter-inspired), with each iteration addressing the limitations of its predecessor.
- **Successful Prototype Realization:** The final mechanism achieved synchronized four-wing flapping at the target frequency of 20 Hz under 7.4 V supply, matching theoretical predictions with $<1\%$ error.
- **Mass Target Compliance:** Through careful material selection and component integration, the total mechanism mass was maintained at 82.4 g (including balance weights), aligning with the <100 g target for the flight-ready vehicle.
- **Rigorous Experimental Validation:** A custom testing framework was established using laser Doppler vibrometry to precisely characterize wingbeat frequency and stability, providing high-fidelity data to validate the dynamic models.

6.2 Main Innovations

This work contributes to the field of bio-inspired robotics through several innovative approaches:

- **Sequential Design Methodology:** Unlike studies focusing on a single optimal solution, this project documented the full failure-analysis cycle of multiple mechanisms. This comparative approach provides valuable insights into the trade-offs between magnetic, direct-drive, and gear-driven actuation for micro-scale flight.
- **Integrated Eccentric Conversion:** The adaptation of the BionicOpter's eccentric shaft concept to a reduced-scale (185 mm wingspan) vehicle demonstrated that complex kinematic transformation is feasible without the mass penalty of traditional crank-rocker linkages.
- **Hybrid Manufacturing Approach:** The combination of precision off-the-shelf components (motors, bearings) with custom 3D-printed structural elements allowed for rapid prototyping without sacrificing kinematic accuracy.
- **Design for Manufacturability:** The detailed design phase, particularly for the Dove and BionicOpter mechanisms, incorporated assembly considerations such as keying, tolerances, and modularity, bridging the gap between conceptual CAD and physical reality.

6.3 Suggestions for Future Works

Building on the validated mechanical platform, the following development paths are recommended to achieve autonomous free flight:

6.3.1 Short-Term Improvements (Mechanical Refinement)

- **Body Structure:** Transition from the current plastic enclosure to a carbon fiber composite shell. This will significantly increase structural rigidity, reduce mass by an estimated 20%, and eliminate flex-induced energy losses.
- **Gear Durability:** Replace the current plastic gears with high-grade POM (Delrin) or brass components to extend operational lifespan and reduce friction and backlash.
- **AoA Control:** Replace the elastic band system with a positive mechanical linkage for rear wing pitch control, ensuring reliable vectoring at high frequencies.

6.3.2 Medium-Term Goals (Flight Control)

- **Flight Controller Integration:** Integrate a 6-axis IMU and flight microcontroller to stabilize the vehicle using the demonstrated differential wing control authority.
- **Closed-Loop Speed Control:** Implement Hall-effect or optical feedback to regulate flapping frequency precisely, compensating for battery voltage drop.

6.3.3 Long-Term Research (Optimization)

- **Aerodynamic Optimization:** Conduct wind tunnel testing to optimize wing flexibility and planform shape for maximum lift-to-drag ratio.
- **Power System:** Investigate high-discharge Li-HV batteries or hybrid capacitor systems to extend flight endurance beyond the current 5-minute estimate.

References

- [1] Z. A. Khan and S. K. Agrawal, "Wing force and moment characterization of flapping wings for micro air vehicle application," University of Delaware, Department of Mechanical Engineering, Tech. Rep. ADA433708, 2005, available via DTIC.
- [2] W. Yang, L. Wang, and B. Song, "Dove: A biomimetic flapping-wing micro air vehicle," *International Journal of Micro Air Vehicles*, vol. 10, no. 1, pp. 70–84, 2018.
- [3] J. W. Gerdes, S. K. Gupta, and S. A. Wilkerson, "A review of bird-inspired flapping wing miniature air vehicle designs," *Journal of Mechanisms and Robotics*, vol. 4, no. 2, pp. 021 003–1–021 003–11, 2012.
- [4] N. Gaissert, R. Mugrauer, G. Mugrauer, A. Jebens, K. Jebens, and E. M. Knubben, "Inventing a micro aerial vehicle inspired by the mechanics of dragonfly flight," in *Towards Autonomous Robotic Systems (TAROS 2013)*, ser. Lecture Notes in Artificial Intelligence, vol. 8069. Springer, 2014, pp. 90–100.
- [5] Q. Liu, Q. Li, X. Zhou, P. Xu, L. Ren, and S. Pan, "Development of a novel flapping wing micro aerial vehicle with elliptical wingtip trajectory," *Mechanical Sciences*, vol. 10, pp. 355–362, 2019.
- [6] E. Pan, L. Chen, B. Zhang, and W. Xu, "A kind of large-sized flapping wing robotic bird: Design and experiments," in *Intelligent Robotics and Applications (ICIRA 2017), Part III*, ser. Lecture Notes in Artificial Intelligence, vol. 10464. Springer, 2017, pp. 538–550.
- [7] P. Lane, G. Throneberry, I. Fernandez, M. Hassanalain, R. Vasconcellos, and A. Abdelkefi, "Towards bio-inspiration, development, and manufacturing of a flapping-wing micro air vehicle," *Drones*, vol. 4, no. 3, p. 39, 2020.
- [8] T. N. Pornsin-Sirirak, Y.-C. Tai, C.-M. Ho, and M. Keennon, "Microbat: A palm-sized electrically powered ornithopter," Paper, 2001, no DOI/venue provided in the supplied original metadata; entry kept non-committal to avoid inventing fields. [Online]. Available: <https://www.semanticscholar.org/paper/Microbat%3A-A-Palm-Sized-Electrically-Powered-Pornsin-Sirirak-Tai/8b7f7d16bd6b629f390503434d9b83c5c56aee65>
- [9] L. Liu, Z. Fang, and Z. He, "Optimization design of flapping mechanism and wings for flapping-wing MAVs," in *Intelligent Robotics and Applications (ICIRA 2008), Part I*, ser. Lecture Notes in Artificial Intelligence, vol. 5314. Springer, 2008, pp. 245–255.

REFERENCES

- [10] T. J. Smith, "Development, design, manufacture and test of flapping wing micro aerial vehicles," Ph.D. dissertation, Wright State University, 2016, Ph.D. dissertation. [Online]. Available: https://corescholar.libraries.wright.edu/etd_all/1695/
- [11] N. T. Jafferis, E. F. Helbling, M. Karpelson, and R. J. Wood, "Untethered flight of an insect-sized flapping-wing microscale aerial vehicle," *Nature*, vol. 570, no. 7762, pp. 491–495, 2019.
- [12] D. Bie, D. Li, J. Xiang, H. Li, Z. Kan, and Y. Sun, "Design, aerodynamic analysis and test flight of a bat-inspired tailless flapping wing unmanned aerial vehicle," *Aerospace Science and Technology*, vol. 112, p. 106557, 2021.
- [13] K. Kakuta, "Flapttter development history," 2023. [Online]. Available: <http://kakutaclinic.life.coocan.jp/DvlpFla.pdf>
- [14] —, "Dragonfly-type ornithopter development history," 2023. [Online]. Available: <http://kakutaclinic.life.coocan.jp/DvlpHDFY.pdf>
- [15] C. Urban, "PteraSoftware," GitHub repository, 2024, accessed: 2024. [Online]. Available: <https://github.com/camUrban/PteraSoftware>
- [16] L. Wang, X. Yang, Y. Luo, H. Chen, and J. Li, "Design of attitude control system for dragonfly-inspired flapping wing aerial vehicle," in *Proceedings of the ICAS 2024 Congress*, 2024, pp. 1–13, ICAS paper 2024_0578 (PDF). [Online]. Available: https://www.icas.org/icas_archive/icas2024/data/papers/icas2024_0578_paper.pdf

Appendix A

Motor Selection Data

This appendix presents the detailed motor selection data discussed in Chapter 5. Table A.1 compares various motors based on their specifications.

A.1 Motor Comparison Table

Table A.1: Motor selection based on MRT

Motor Name	Mass (grams)	Kv	Kt	RPM (No Load)			I %50 (A)			I %100 (A)			Torque (N.m)					
				1S	2S	3S	1S	2S	3S	1S	2S	3S	I %50 (A)			I %100 (A)		
													1S	2S	3S	1S	2S	3S
RCinpower GTS V2 1202.5	3.9 ~ 4.3	6000	0.0016	22200	44400	66600	0	1.2	1.3	0	4.2	5.17	0	0.0019	0.0020	0	0.0067	0.0082
		11500	0.0008	42550	85100	127650	1.86	2.95	0	6.87	10.8	0	0.0015	0.0024	0	0.0057	0.0090	0
RCinpower GTS V3 1203	4.5	6000	0.0016	22200	44400	66600	0	1.3	1.45	0	4.55	5.7	0	0.0021	0.0023	0	0.0072	0.0091
		8000	0.0012	29600	59200	88800	0	1.97	3.15	0	6.64	8.9	0	0.0024	0.0038	0	0.0079	0.0106
		11500	0.0008	42550	85100	127650	2.05	3.25	0	7.55	11.9	0	0.0017	0.0027	0	0.0063	0.0099	0
T-motor M1104 / M1106	7.24	6000	0.0016	22200	44400	66600	0	0	12.6	0	0	14.6	0	0	0.0200	0	0	0.0232
	5.6	7500	0.0013	27750	55500	83250	0	3.4	3.4	0	12.7	13.1	0	0.0043	0.0043	0	0.0162	0.0167
T-motor F1203	5.25	7000	0.0014	25900	51800	77700	0	0	2.82	0	0	8.67	0	0	0.0038	0	0	0.0118
T-motor M1103	4.64	8000	0.0012	29600	59200	88800	0	0	2.2	0	0	7.4	0	0	0.0026	0	0	0.0088
EMAX TH1103	4.1	7000	0.0014	25900	51800	77700	?	0	0	?	0	0	#Val	0	0	#Val	0	0
HGLRC AEOLUS 1202.5	~ 4.6	5600	0.0017	20720	41440	62160	0	1.1	1.23	0	4.05	5.07	0	0.0019	0.0021	0	0.0069	0.0086
		8000	0.0012	29600	59200	88800	0	1.36	1.9	0	5.53	6.82	0	0.0016	0.0023	0	0.0066	0.0081
F15 KV6000	7.8	6000	0.0016	22200	44400	66600	0	1.95	2.8	0	8.02	10.1	0	0.0031	0.0044	0	0.0128	0.0161